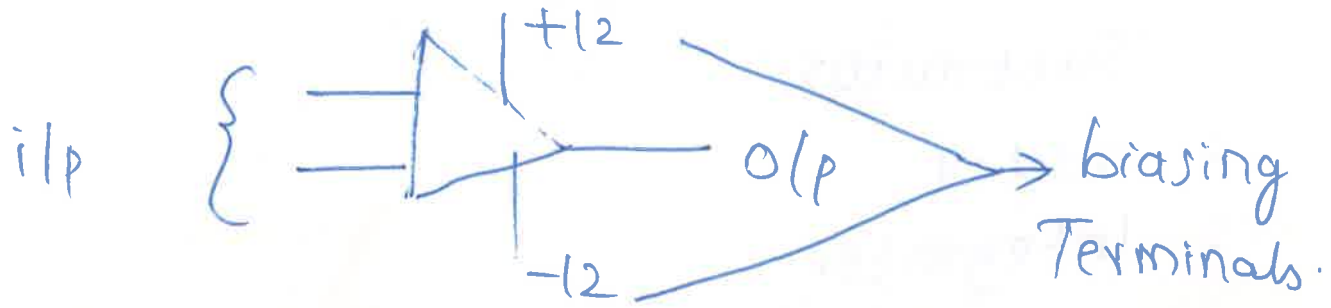
# Operational Amplifiers:

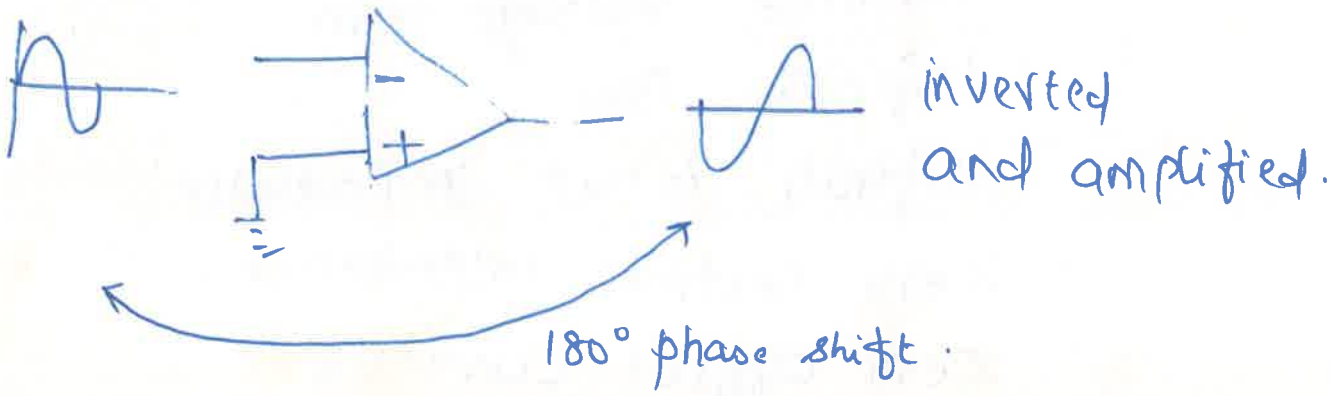
- It is an amplifier consisting of many number of transistors.
- Operational amplifiers are shortly called as OPAMPs.
- The symbol of OPAMP is,



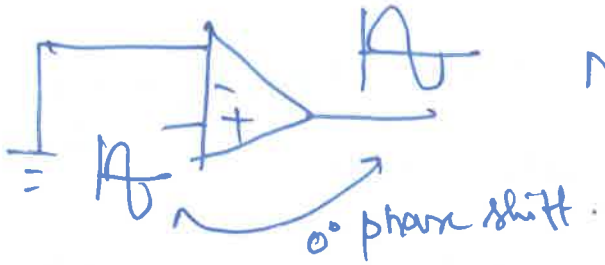

## Inputs:

There are two kinds of inputs.
- inverting input
- non inverting input.

### Inverting input:



## Non inverting input:




non inverted ~~o/p~~ and amplified output.

## Application of OPAMP:

- As an amplifier
- A Buffer element.
- Subtractor.
- adder
- Integrator
- differentiator.

Most mathematical operations are possible and hence they are known as operational amplifier.

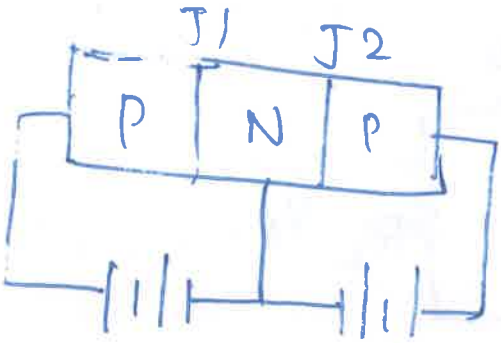
## Characteristics of Ideal OPAMP:

- Infinite Voltage gain.
- Infinite BW
- Infinite input impedance
- Zero output impedance.
- Zero Offset Current.

# Biasing an OPAMP:

- OPAMP consists of many transistors.
- Transistors must be made to operate as amplifier by biasing the V & I of transistors to operate in active region.

- As switch in ON condition - Saturation region.
- " " " OFF " " - Cutoff region.




~~Junctions biasing application~~

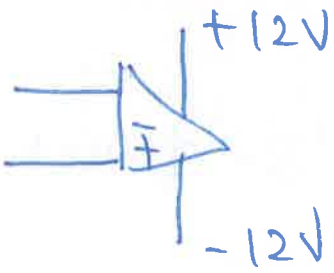
~~$J_1$ $J_2$ FB~~

~~$J_2$ RB~~

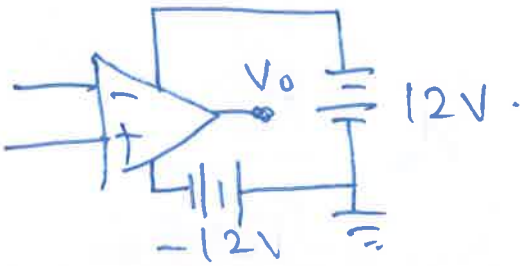
~~} Amplifier~~

| Biasing | $J_1$ | $J_2$ | application. |
|---|---|---|---|
| | FB | RB | amplifier. |
| | FB | FB | S/w-ON |
| | RB | RB | Cutoff. |

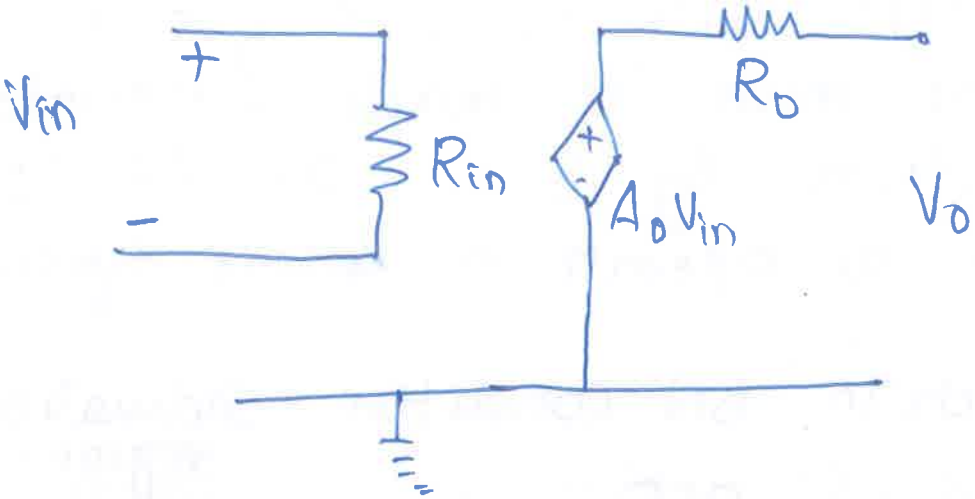
## OPAMP Model:



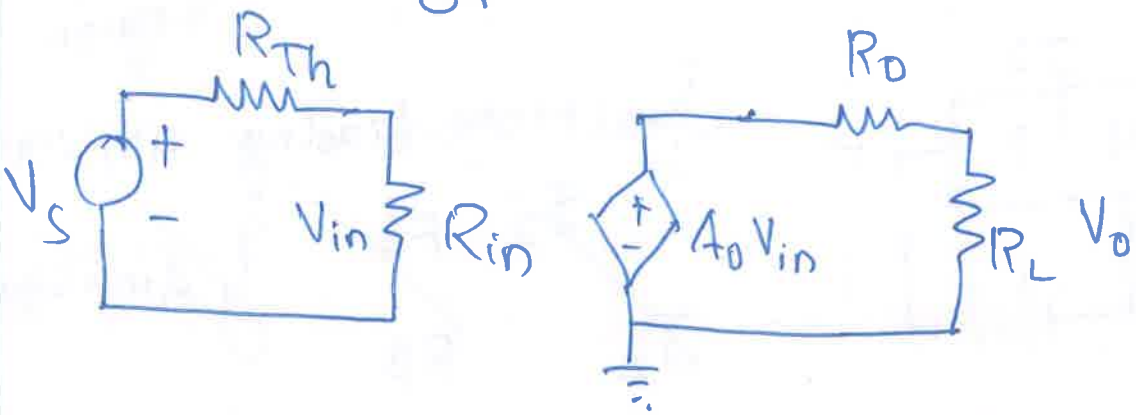

⇒



# OPAMP equivalent circuit model:-




or.



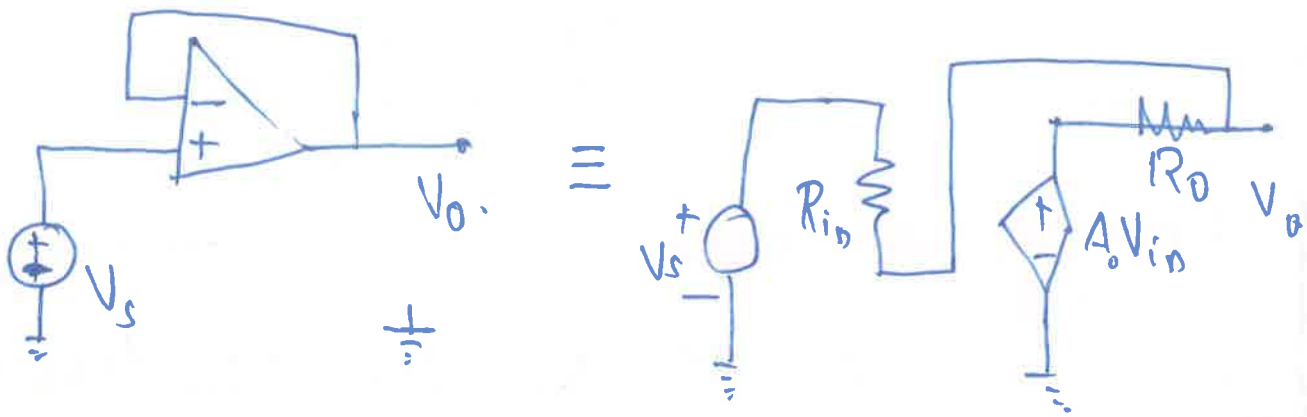

$$V_{in} = V_s \frac{R_{in}}{R_{in}+R_{Th}}. \qquad V_o = A_o V_{in} \frac{R_L}{R_L+R_o}.$$

$$V_o = A_o V_s \frac{R_{in}}{R_{in}+R_{Th}} \times \frac{R_L}{R_L+R_o}.$$

$\frac{V_o}{V_s}$ = Transfer function or gain.

$$\frac{V_o}{V_s} = \frac{R_{in}}{R_{in}+R_{Th}} \cdot A_o \cdot \frac{R_L}{R_L+R_o}.$$

# Buffer circuit or unit gain buffer:-




KVL ⇒ $V_s = I R_{in} + I R_o + A_o V_{in}$.

$V_o = I R_o + A_o V_{in}$.

$V_{in} = I R_{in}$.

$$\frac{V_o}{V_s} = \frac{I R_o + A_o V_{in}}{I R_o + A_o V_{in} + I R_{in}}.$$

$$= \frac{R_o + A_o R_{in}}{R_o + A_o R_{in} + R_{in}}.$$

$$= \frac{1}{1 + \frac{R_{in}}{R_o + A_o R_{in}}} \longrightarrow \frac{1}{\frac{R_o}{R_{in}} + A_o}.$$

$R_{in} <<< R_o$.

$$V_o/V_s \approx \frac{1}{1 + 1/A_o}.$$

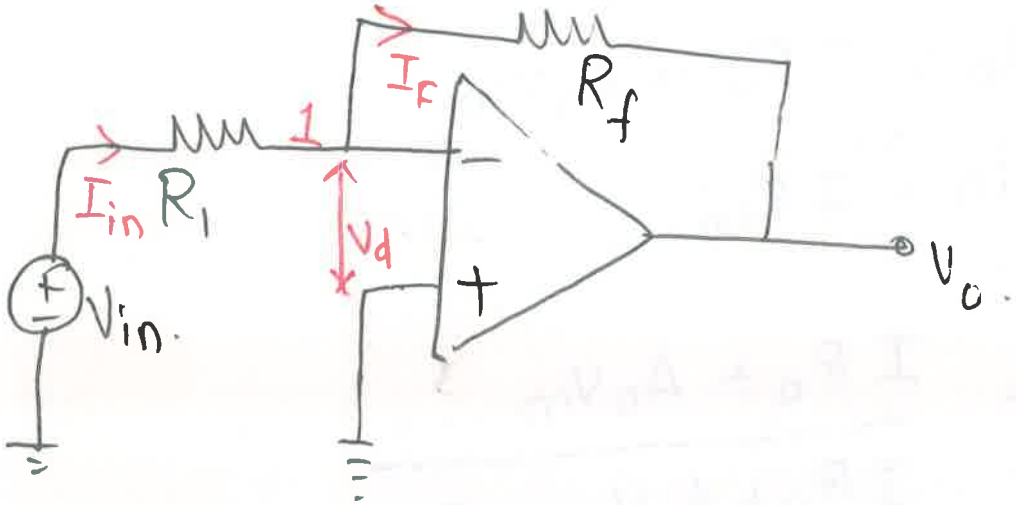
If $A_0 >> 1$

$$\frac{V_0}{V_s} = 1 \Rightarrow \text{unity gain Amplifier.}$$

## Inverting ~~an~~ OPAMP:

For an opamp, the differential Voltage $V_d = V^+ - V^- = 0$.

Therefore $V^+ = V^-$.




Applying KCL at 1,

$$I_{in} = I_F \quad [\because \text{No current enters OPAMP due to high } Z_{in}].$$

$$\frac{-V_1 + V_{in}}{R_1} = \frac{V_1 - V_0}{R_F}.$$

$$\frac{V_1}{R_1} - \frac{V_{in}}{R_1} = \frac{V_1}{R_F} - \frac{V_0}{R_F}$$

$$\frac{V_{in}}{R_1} = -\frac{V_o}{R_F}.$$

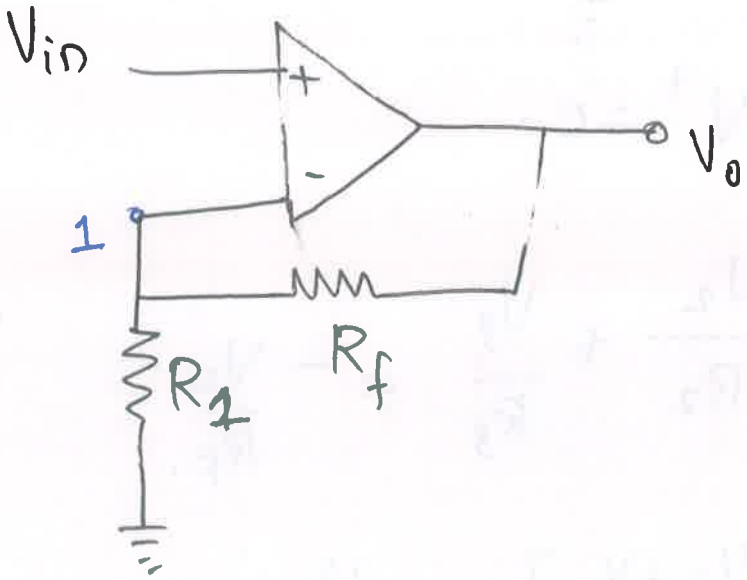
$V_1 = 0.$

$$\frac{V_o}{V_{in}} = -\frac{R_F}{R_{in}}.$$

Gain $= \frac{R_F}{R_1}$ ; $V_o = -A_o V_{in}$ → o/p is inverted

Hence 180° phase shift.

## Non inverting OPAMP:-



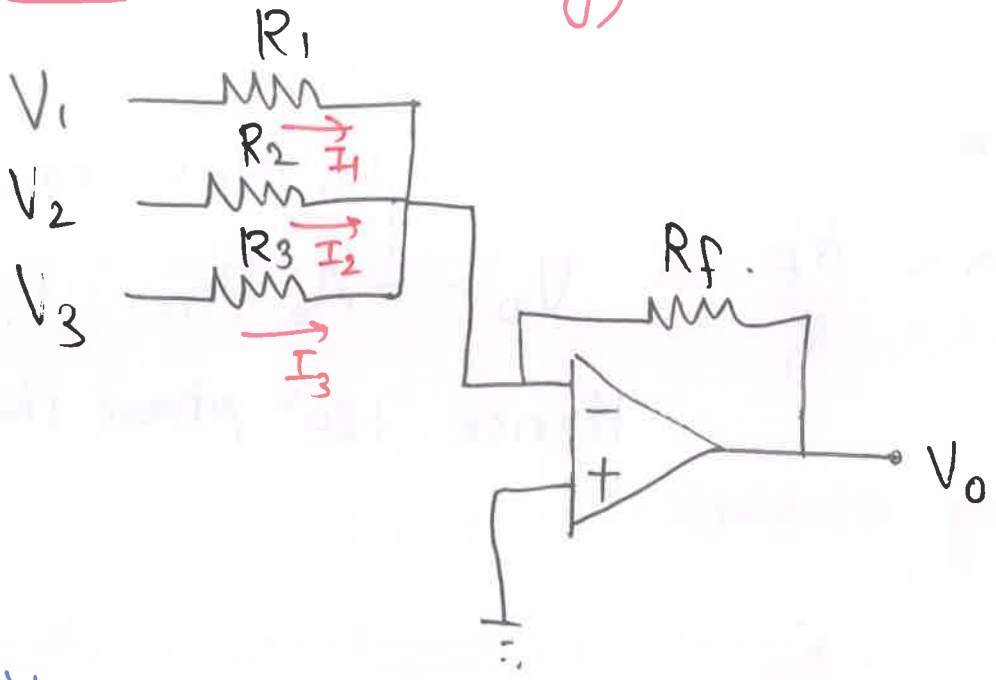

$V_d = 0 \quad V^+ = V^- = V_{in}.$

$$V_{R_1} = V_{in} = V_{out} \cdot \frac{R_1}{R_1 + R_f}.$$

$$\therefore \frac{V_o}{V_{in}} = \frac{R_1 + R_f}{R_1} = 1 + \frac{R_F}{R_1}$$

$$\frac{V_o}{V_{in}} = A_o \quad , \text{ Where } A_o = 1 + \frac{R_f}{R_1}$$

## Summer: (Inverting)




$$V_d = V^- - V^+ = 0.$$

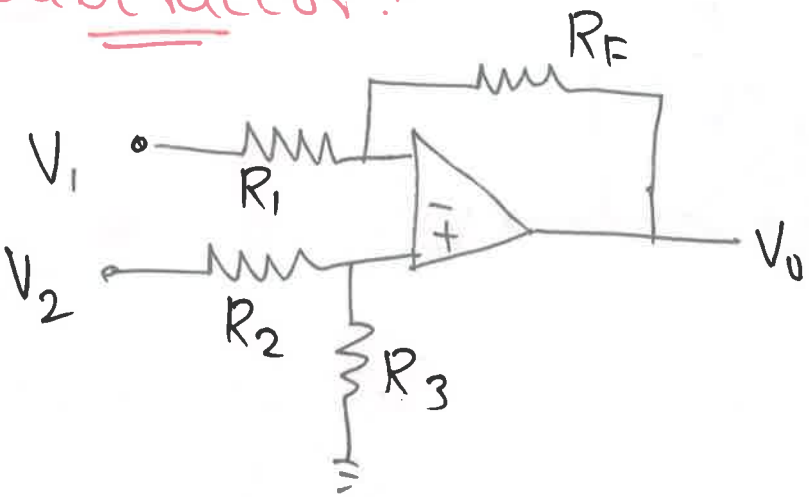
$$\frac{V_1 - 0}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} = -\frac{V_o}{R_F}.$$

$$\frac{1}{R}\left[V_1 + V_2 + V_3\right] = -\frac{V_o}{R_F} \quad [\text{IF } R_1 = R_2 = R_3].$$

$$V_o = -\left[V_1 + V_2 + V_3\right] \quad [\text{IF } R_F = R].$$

Summer.

# Subtractor :-




$$V_d = 0 \Rightarrow V^- = V^+.$$

$$V^+ = V_2 \frac{R_3}{R_3+R_2} = V^-$$

$$\frac{V_1 - V^-}{R_1} = \frac{V^- - V_0}{R_F}.$$

$$\frac{V_1}{R_1} - \frac{1}{R_1}\left[V_2 \frac{R_3}{R_3+R_2}\right] = \frac{1}{R_F}\left[V_2 \frac{R_3}{R_3+R_2}\right] - \frac{V_0}{R_F}.$$

$$\frac{V_1}{R} - \frac{V_2}{2R} = \frac{V_2}{2R} - \frac{V_0}{R}.$$

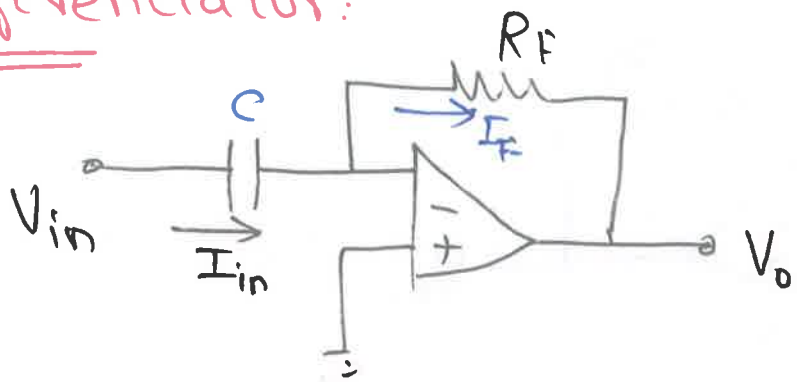
$$V_0 = \frac{V_1}{R} - \frac{V_2}{2R} - \frac{V_2}{2R}.$$

$$= \frac{1}{R}\left[V_1 - V_2(1)\right] = \boxed{\frac{1}{R}(V_1 - V_2) = V_0}$$

IF $R_2 = R_3 = ?$.

$$V_0 = \frac{R_F}{R_1}(V_1 - V_2)$$

## Differentiator:



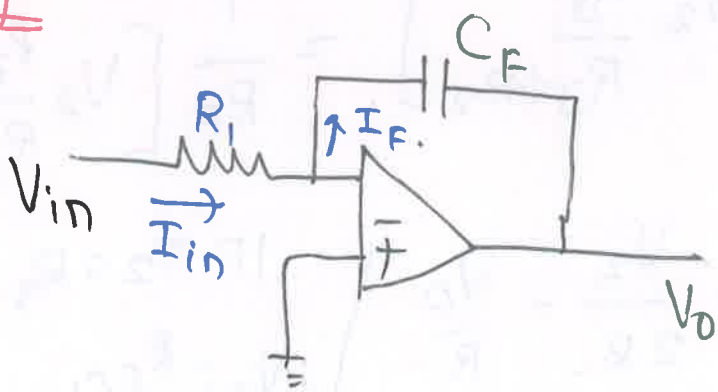

$$I_{in} = I_F.$$

$$C\frac{d[V_{in} - V^-]}{dt} = \frac{V^- - V_o}{R_F}$$

$$\cancel{V_o = \frac{-RC}{R_F}\frac{dV_{in}}{dt}} \qquad C\frac{dV_{in}}{dt} = -\frac{V_o}{R_F} \qquad [V^- = V^+ = 0]$$

$$\boxed{V_o = -R_F C \frac{dV_{in}}{dt}.}$$

## Integrator:



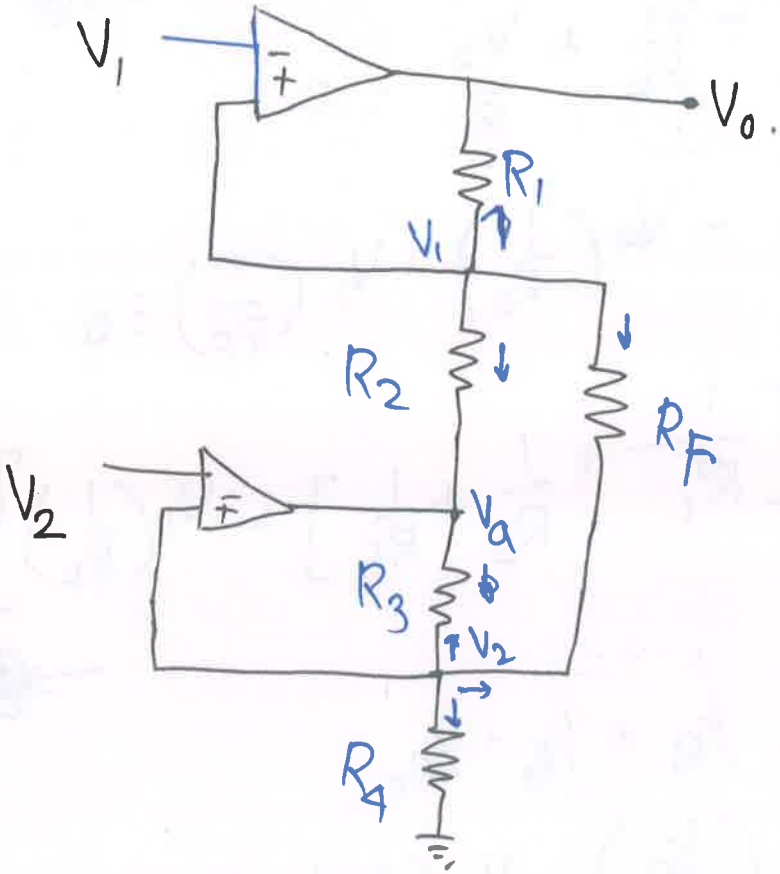

$$I_{in} = I_F$$

$$\frac{V_{in} - V^-}{R_1} = C\frac{d[V^- - V_o]}{dt}$$

$$\frac{V_{in}}{R_1} = -C\frac{dV_o}{dt}.$$

$$\boxed{V_o = \frac{-1}{R_1 C}\int V_{in}\,dt.}$$

## Instrumentation amplifier:-




$$\frac{V_o - V_1}{R_1} + \frac{V_1 - V_a}{R_2} + \frac{V_1 - V_2}{R_F} = 0 \longrightarrow (1)$$

$$\frac{V_2 - V_a}{R_3} + \frac{V_2 - V_1}{R_F} + \frac{V_2}{R_4} = 0 \longrightarrow (2)$$

let $R_2 = R_3$ and $R_1 = R_4$.

① ⇒ $$\frac{V_1}{R_1} - \frac{V_a}{R_1} + \frac{V_1}{R_2} - \frac{V_a}{R_2} + \frac{V_1}{R_F} - \frac{V_2}{R_F} = 0.$$

$$V_1\left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_F}\right] - V_a\left(\frac{1}{R_2}\right) - V_0\left[\frac{1}{R_1}\right] - V_2\left[\frac{1}{R_F}\right] = 0. \rightarrow ③$$

② ⇒ $$\frac{V_2}{R_2} - \frac{V_a}{R_2} + \frac{V_2}{R_F} - \frac{V_1}{R_F} + \frac{V_2}{R_1} = 0$$

$$V_2\left[\frac{1}{R_2} + \frac{1}{R_F} + \frac{1}{R_1}\right] - V_a\left(\frac{1}{R_2}\right) - V_1\left(\frac{1}{R_F}\right) = 0$$

$$V_a = R_2\left\{V_2\left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_F}\right] - V_1\left(\frac{1}{R_F}\right)\right\} \rightarrow ④$$

④ in ③ ⇒

$$\delta = 1/R_1 + 1/R_2 + 1/R_F.$$

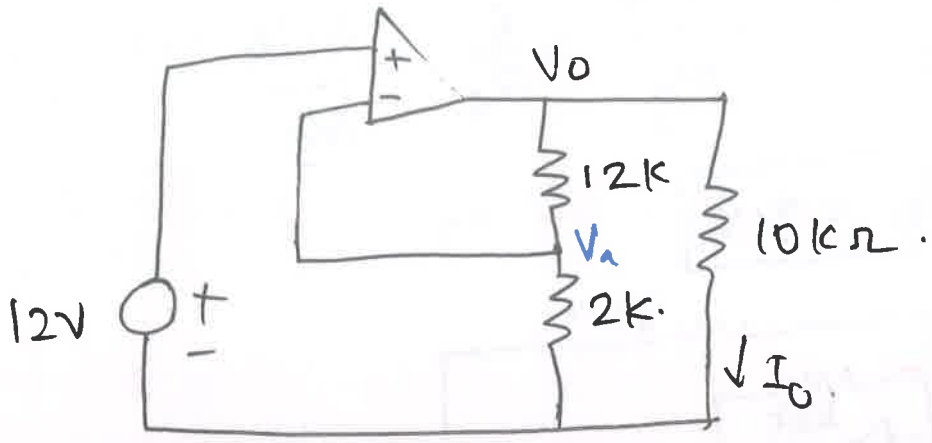
$$V_1[\delta] - V_2[\delta] + V_1(1/R_F) - V_0(1/R_1) - V_2(1/R_F) = 0.$$

$$V_1[\Delta] - V_2[\Delta] - V_0[1/R_1] = 0.$$

$$\boxed{V_0 = [V_1 - V_2]\left[1 + \frac{R_1}{R_2} + 2\frac{R_1}{R_F}\right]}$$

eg:107

$I_o = ?$




$V_{2K} = 12V$

$\cancel{V_{10K} = 12 \times \frac{10K}{12K}} =$

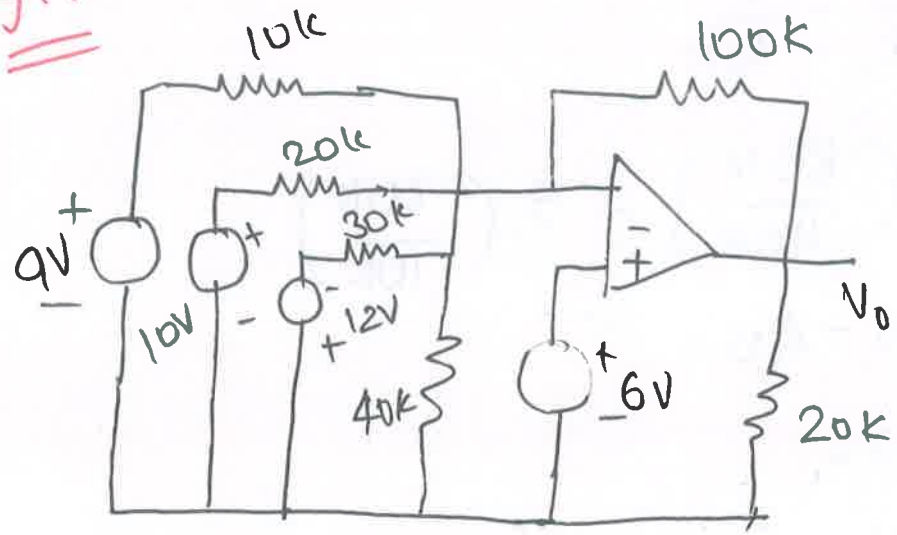
$$\frac{V_a}{2K} + \frac{V_a - V_o}{12K} = 0.$$

$$V_a\left[\frac{1}{2K} + \frac{1}{12K}\right] = \frac{V_o}{12K}.$$

$$V_o = -84V.$$

$$I_o = \frac{V_o}{10K} = 8.4mA.$$

eg:108

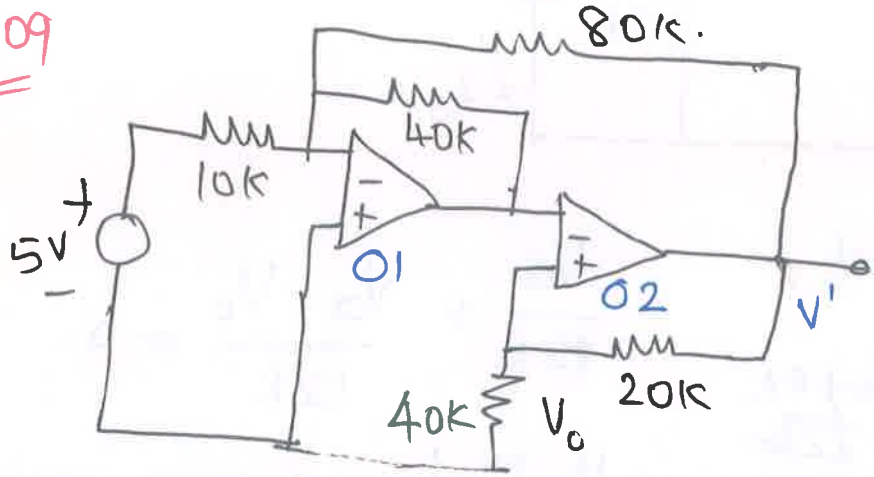


$V^+ = V^- = 6V.$

$$\frac{9-6}{R_1} + \frac{10-6}{R_2} + \frac{-12-6}{R_3} = \frac{6}{R_4} + \frac{6-V_0}{R_5}$$

$$\boxed{V_0 = 31V.}$$

eg: 109




KCL at $V^-$ of O1:

$$\frac{5-0}{10k} = \frac{0-V_0}{40k} + \frac{0-V'}{80k.}$$

$$\frac{V'}{80k} = \frac{-V_0}{40k} - \frac{5}{10k}$$

$$V' = -V_0\left(\frac{80k}{40k}\right) - 5\left(\frac{80k}{10k}\right)$$

$$V' = -2V_0 - 40$$

KCL at $V^+$ of O2:

$$\frac{V_0}{40k} + \frac{V_0 - V'}{20k} = 0$$

$$V' = \left[\frac{V_0}{40K} + \frac{V_0}{20K}\right] 20K.$$

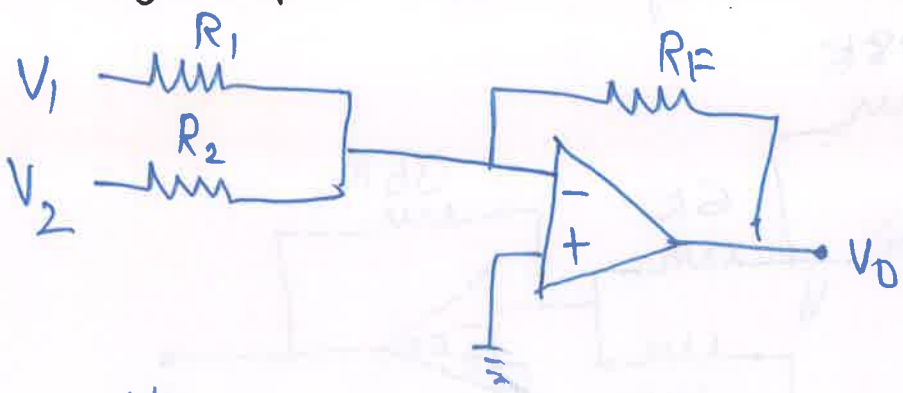
$$= V_0\left(\frac{1}{2}\right) + V_0 = \frac{1.5}{3} V_0.$$

$$1.5V_0 = -2V_0 - 40$$

$$3.5V_0 = -40$$

$$\boxed{V_0 = -11.43V.}$$

eg:10 Given a box of 10k resistors and an opamp, design a circuit that will have an o/p voltage of $V_0 = -2V_1 - 4V_2$




$$V_0 = -\left(\frac{R_F}{R_1}\right)V_1 - \left(\frac{R_F}{R_2}\right)V_2 \qquad \frac{V_1}{R_1} + \frac{V_2}{R_2} = \frac{-V_0}{R_F}$$

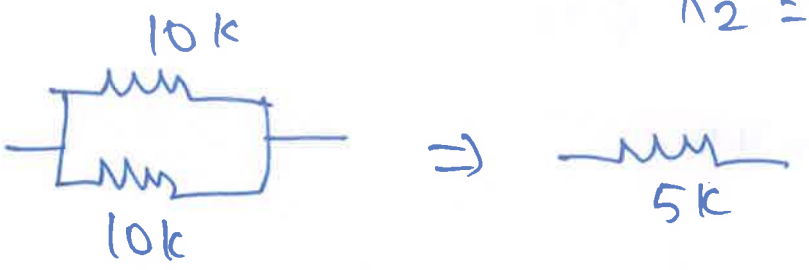
$$V_0 = -2V_1 - 4V_2$$

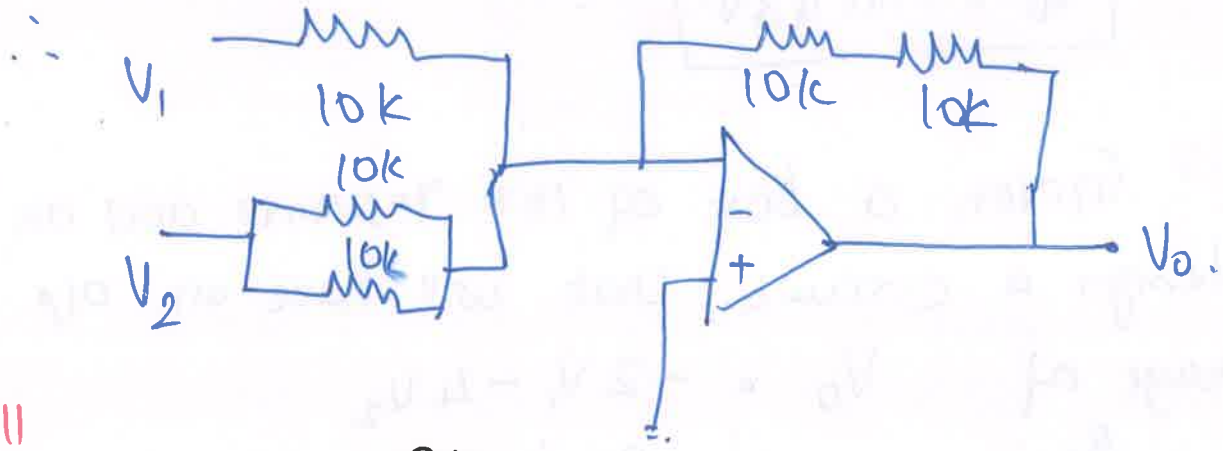
let $R_F = 20k$ i.e., 10k + 10k in series.

$R_1$ will be 10k.

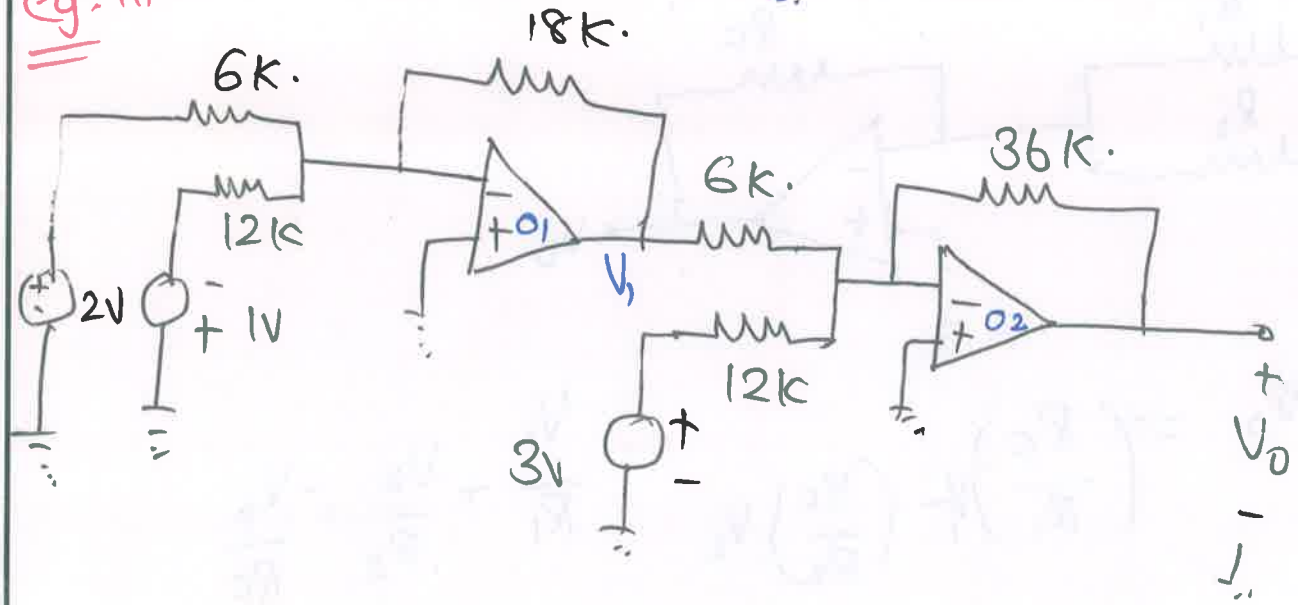
$$\frac{R_F}{R_2} = 4 \Rightarrow \frac{20k}{4} = \cancel{5k}\ R_2.$$

$$R_2 = 5k.$$




eg: III




$V_0 = ?$.

For O1:

$V^+ = V^- = 0$

$$\frac{2}{6k} - \frac{1}{12k} = -\frac{V_1}{18k}.$$

$$V_1 = -2\left(\frac{18k}{6k}\right) + \left(\frac{18k}{12k}\right) = -4.5V.$$

For O2:

$V_o = V^+ = V^- = 0$

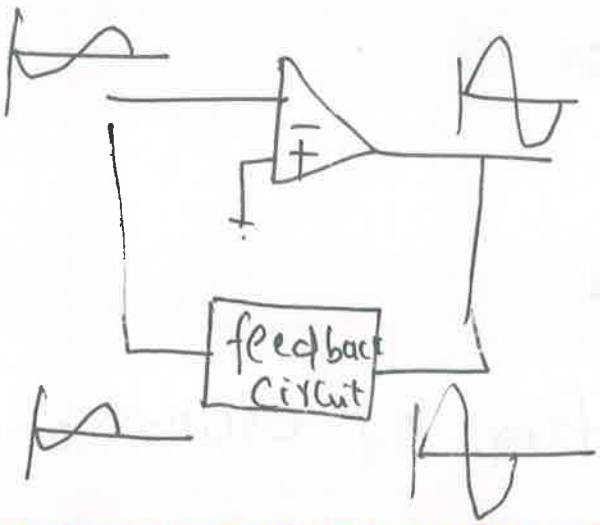
$$-\frac{4.5}{6k} \times 36k + 3\left(\frac{36k}{12k}\right) = -V_0$$

$$\boxed{V_0 = 18V.}$$

## Oscillators - Introduction:

Circuit that generate sinusoidal input without any input signal are called as oscillators.

- They work on the principle of positive feedback.

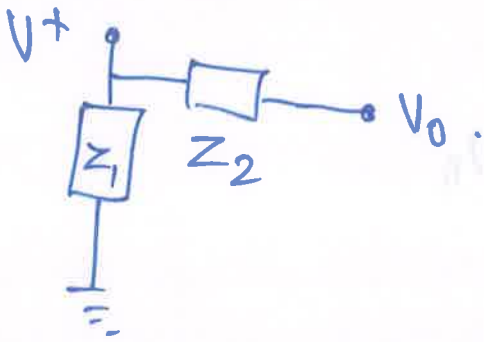



Refer (ppt)

$$V^+ = V^-$$

$$Z_1 = \left(R_3 \parallel \frac{1}{-jX_{C_2}}\right) = \frac{(R_3)\left(\frac{+1}{-jX_{C_2}}\right)}{R_3 - \frac{1}{jX_{C_2}}}$$

$$Z_2 = R_4 - jX_{C_4}.$$




$$V_o = V^+ \frac{Z_1}{Z_1 + Z_2}.$$

$$\frac{V_o}{V^+} = \frac{-(R_3)\left(\frac{1}{jX_{C_2}}\right)}{R_3 - \frac{1}{jX_{C_2}}}$$

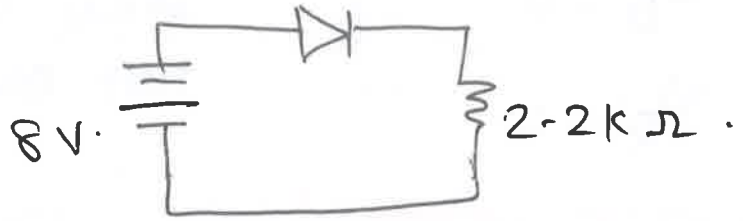
Solving the above for $\omega$

$$\omega = \frac{1}{\sqrt{R_3 R_4 C_p C_2}}.$$

$$\boxed{f = \frac{1}{2\pi RC}}$$ freq. of oscillation.

# ~~Clippers~~: Series diode configuration.

eg: 112: Determine $V_D$, $V_R$ and $I_D$. Assume diode to be a practical Silicon diode.




For non ideal Si diode, $V_D = 0.7V$

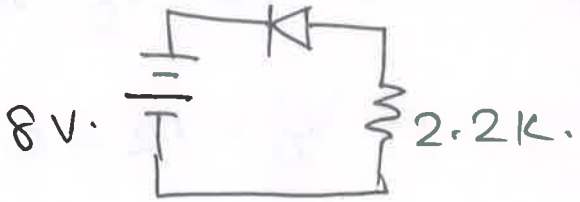
$\cancel{V_R} =$

$$8 - V_D - V_R = 0.$$

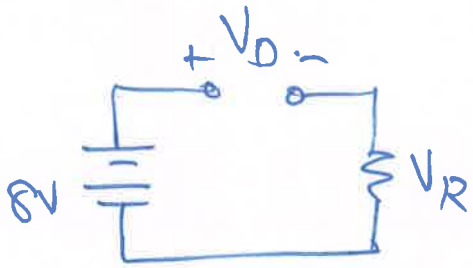
$$V_R = 8 - 0.7 = 7.3V.$$

$$I_D = I_R = \frac{V_R}{2.2k} = \underline{\underline{3.32mA}}$$

eg: 113 Repeat 112 with diode reversed.




Diode is RB.

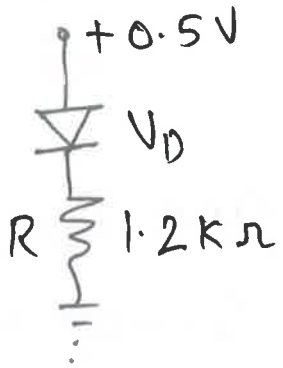



$I_D = I_R = 0$

$\Rightarrow V_R = 0.$

$8 - V_D - V_R = 0. \Rightarrow \boxed{V_D = 8V.}$

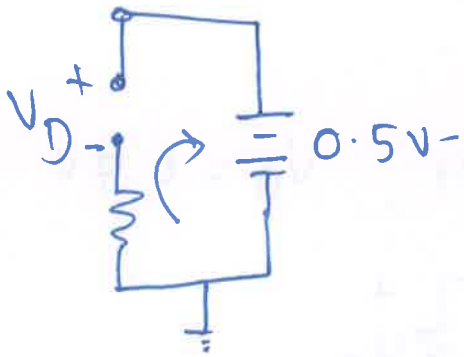
eg: 114




$V_D = ?$, $V_R = ?$ $I_D = ?$

$I_D = 0$ [∵ Diode is still not ON].

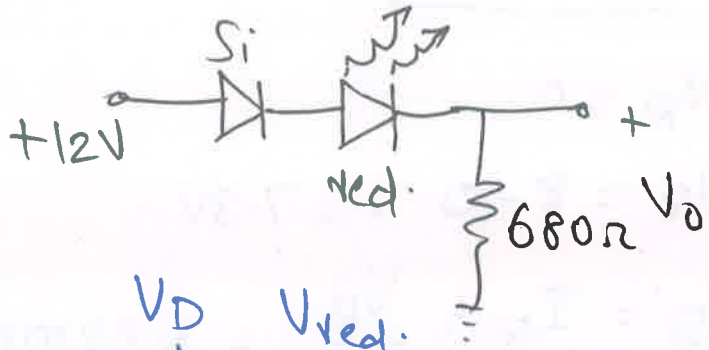
$V_R = 0$.




$V_D - 0.5V = 0$.

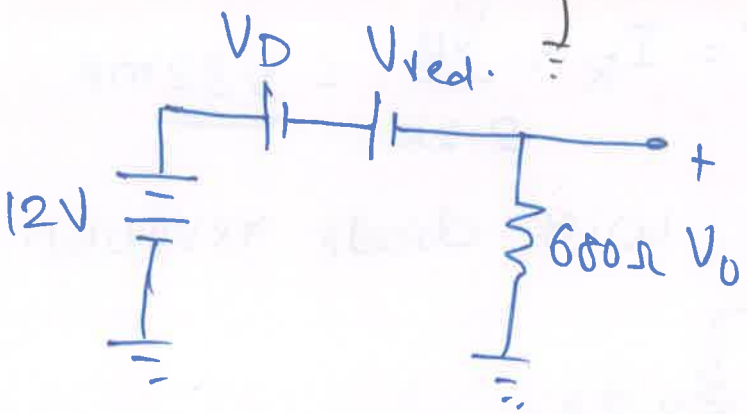
$\boxed{V_D = 0.5V}$.

eg: 115




let $V_{red}$ be 1.8V.

find $V_o$ and $I_D$.



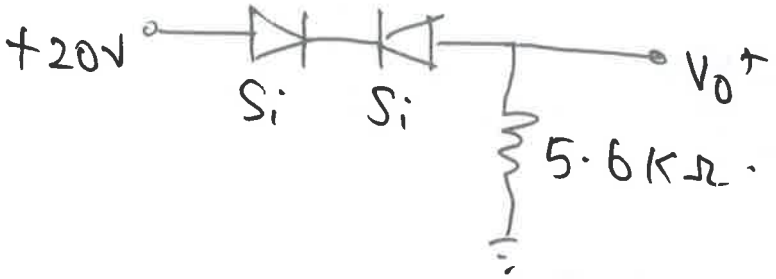

$12 - V_D - V_{red} - V_o = 0$.

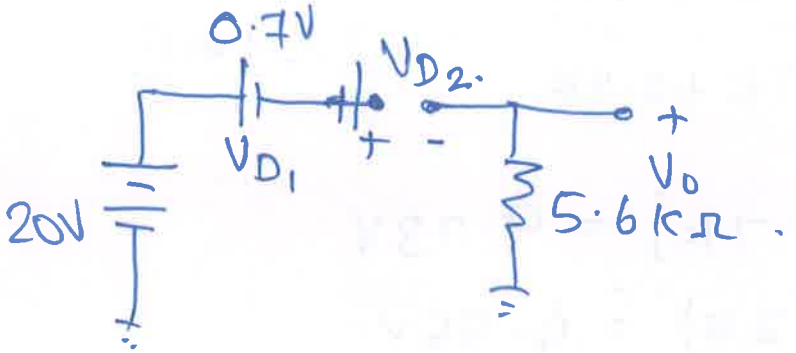
$V_o = 12 - 0.7 - 1.8$

$V_o = 12 - 2.5 = 9.5V$.

$I_D = I_R = \frac{9.5}{680\Omega} = \underline{13.97mA}$

eg:116: Determine $I_D$, $V_{D_2}$ and $V_0$




$I_D = 0$.

$\Rightarrow V_0 = 0$.

$20 - 0.7 - V_{D_2} = 0$. X [wrong. For No biar condition

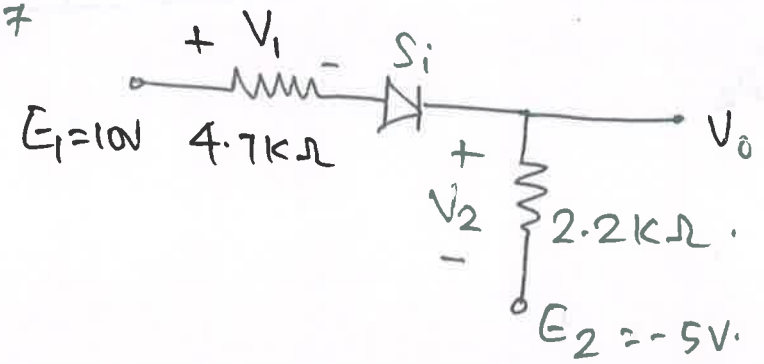
~~$V_{D_2}$~~ :

whenever, $I_D = 0$, always consider $V_0 = 0$ henceforth. Therefore,

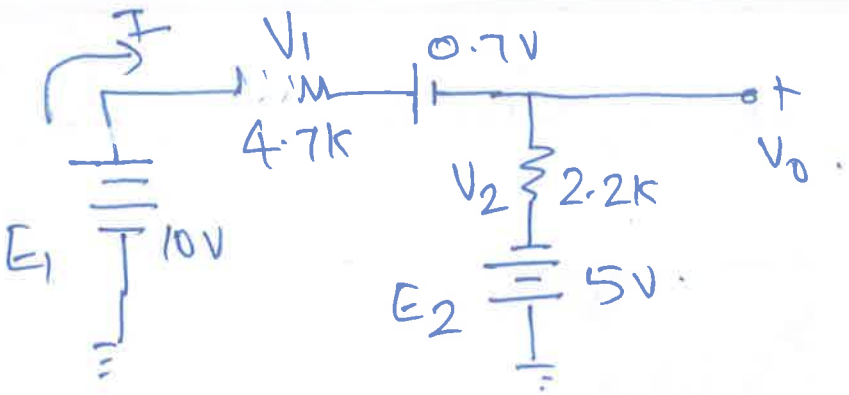
$20 - 0 - V_{D_2} = 0$

$$\boxed{V_{D_2} = 20V}$$

eg:117




$I$, $V_1$, $V_2$, $V_0$.




$$I = \frac{10 - 0.7 + 5V}{4.7K + 2.2K} = 2.07\,mA.$$

$$V_1 = I(4.7k) = 9.73V.$$

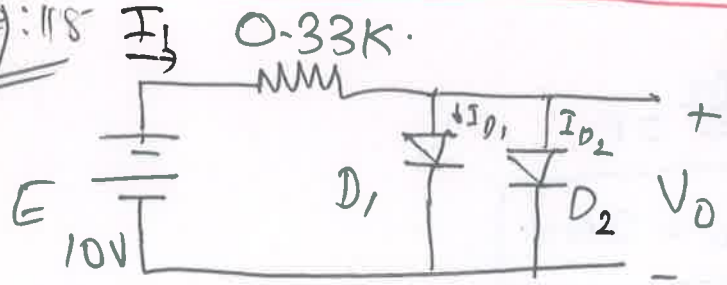
$$V_2 = I(2.2k) = 4.55V.$$

$$-E_2 + V_2 - V_0 = 0.$$

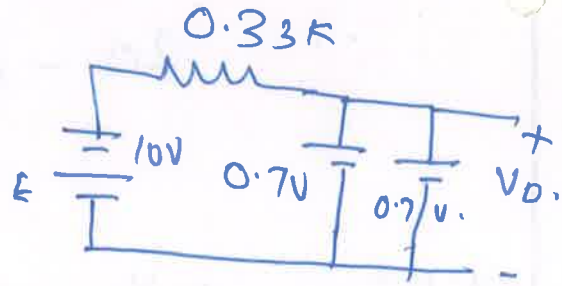
$$V_0 = 4.55 - 5 = -0.45V.$$

## Parallel and Series-parallel Configurations:-

Eg: 1.18




$\equiv$



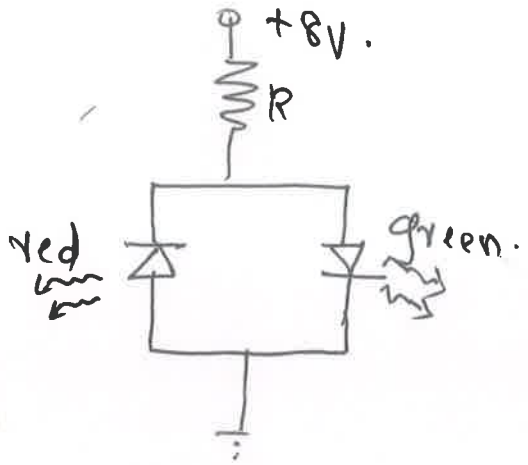

$V_0$, $I_1$, $I_{D_1}$, $I_{D_2}$.

$$V_0 = 0.7V.$$

$$I_1 = \frac{V_R}{0.33K} = \frac{10 - V_{D_1}}{0.33K} = \frac{10 - 0.7V}{0.33K} = 28.18\,mA.$$

$I_{D_1} = I_{D_2} = I_1/2 = \frac{28.18mA}{2} = 14.09mA.$

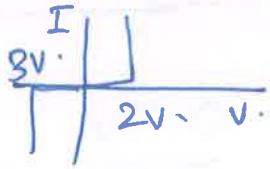
Eg: 119: A polarity detector circuit.

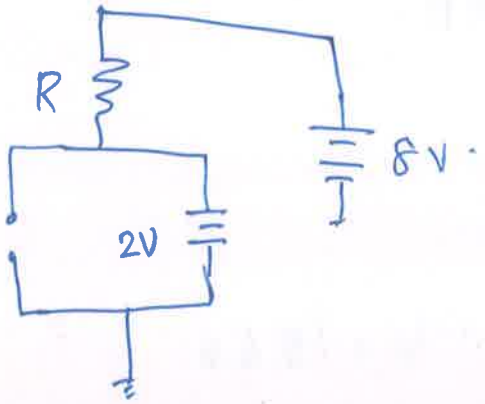



Find R to send 20mA of current.

$D_{ON}$ Voltage = 2V

RBD " = 3V.



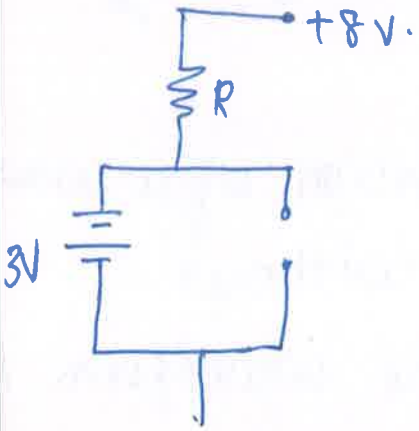




$I = \frac{8-2}{R}$

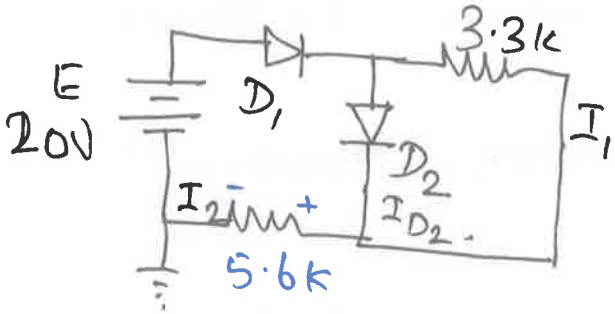
$R = \frac{6}{20mA} = 300\Omega.$

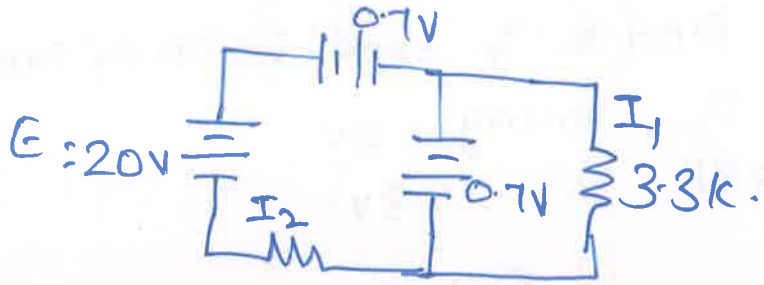
$V_{BR}$ for RED = 2V – OK ✓.

If green LED is replaced with blue with $V_{BR} = 5V$



Eg: 120. $I_1, I_2, I_{D_2}$.




$$I_1 = \frac{0.7V}{3.3k} = 0.212mA.$$

$$E - 0.7V - 0.7V - V_{R_2} = 0.$$

$$V_{R_2} = 20 - 1.4 = 18.6V.$$

$$I_2 = \frac{V_{R_2}}{5.6k} = 3.32mA$$

$$I_{D_2} = I_2 - I_1 = 3.32 - 0.212 = 3.11mA$$

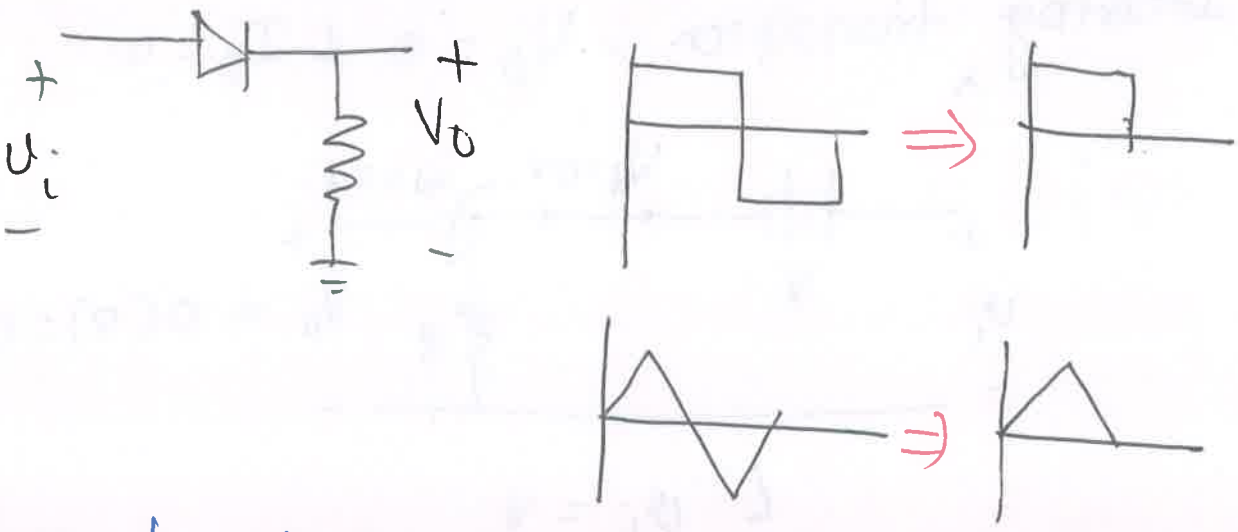
## Clippers:

A n/w that clips a portion of a waveform with a help of diode without altering the other part of the waveform is called clippers.

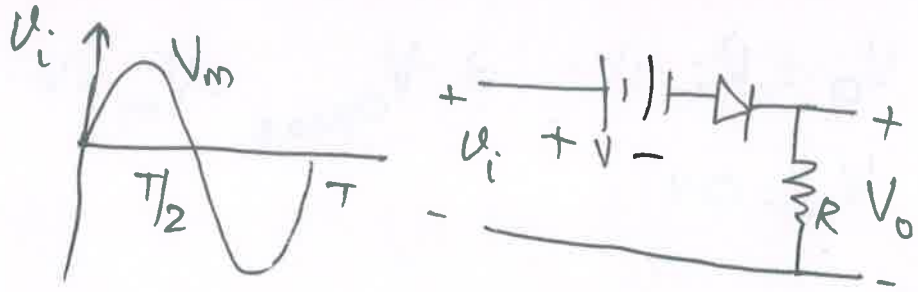
They are classified as
→ Series clipper
→ Parallel clipper.

## a) Series clipper:-

If the diode is in series with the load, then they are called as series clipper.




## Steps to determine the output of the clipper circuit:-




① Where is o/p voltage $V_0$ defined?
— Current flow will not get $V_0$.

② Pressure exerted by each source and the overall effect on diode ON/OFF.

## 3 Determine the transition Voltage.

**Step 2:**

$$V_i - V - V_0 = 0.$$

$$V_0 = V_i - V$$

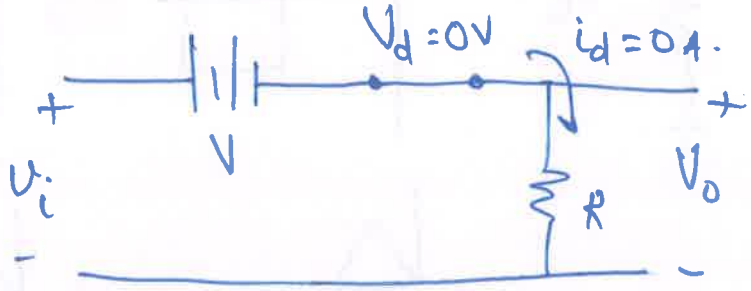
[Diode – ON when $v_i > V$.

**Step 3:**

At the start of ~~During~~ transition, Transition is from OFF TO ON.

$V_D = 0$ & $I_D = 0$.

$\therefore$



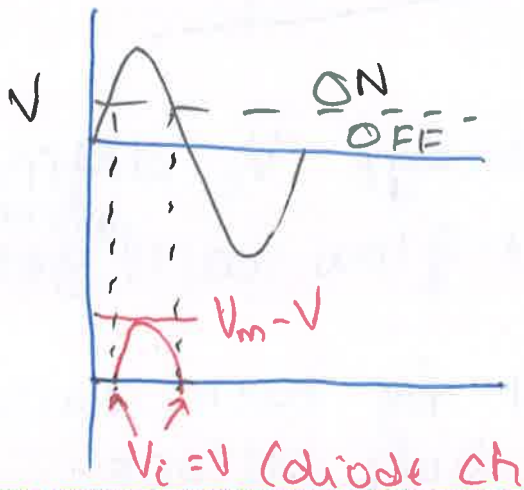

$V_0 = 0(R) = 0$

& $v_i = V$.

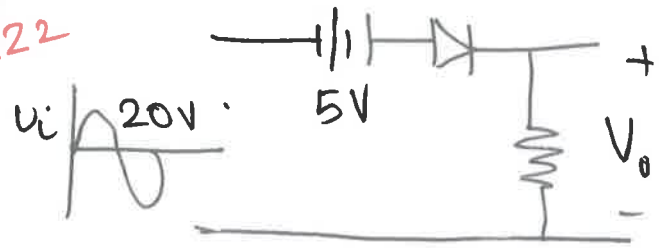
$\Rightarrow$ Transition takes place when $v_i = V$.

once ON, $V_0 = v_i - V$. $\Rightarrow V_{0peak} = V_m - V$.

For OFF, $V_0 = 0V$.



eg: 122



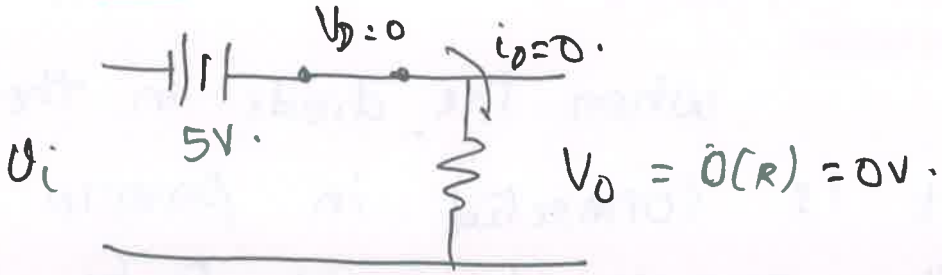

S1: We get $V_o$ whenever diode is ON.

S2: +ve voltage of $V_i$ - FB diode.

+5V - FB diode.

Whenever $-v_i > 5V$ - diode is RB.

S3: Transition model:




$v_i + 5V = 0$

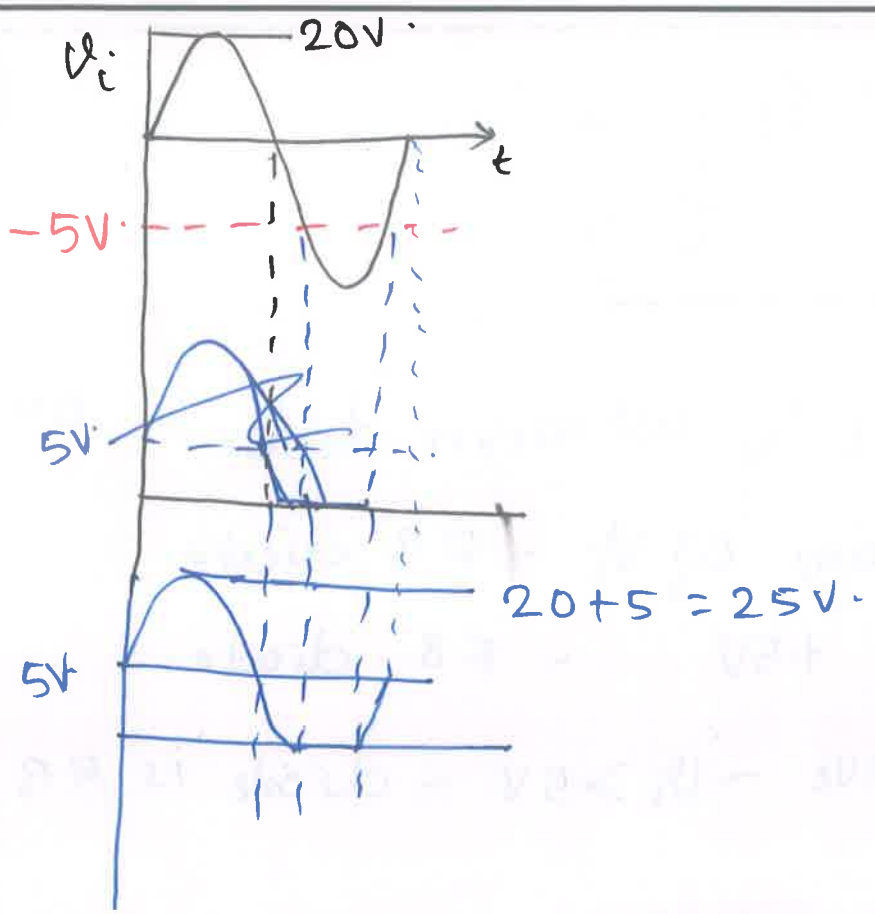
$v_i = -5V$. Transition at $-5V$.

S4: once ON

$v_i + 5 - V_o = 0$.

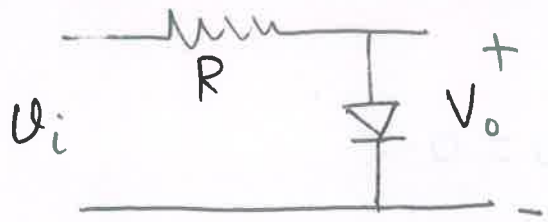
$V_o = v_i + 5$

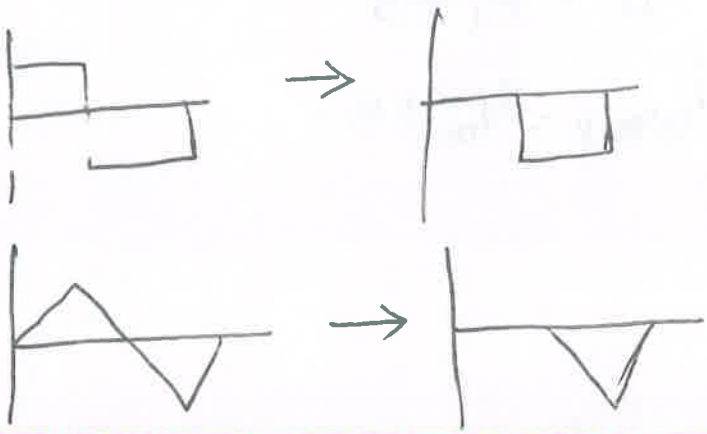
$V_{o\,peak} = V_m + 5$.



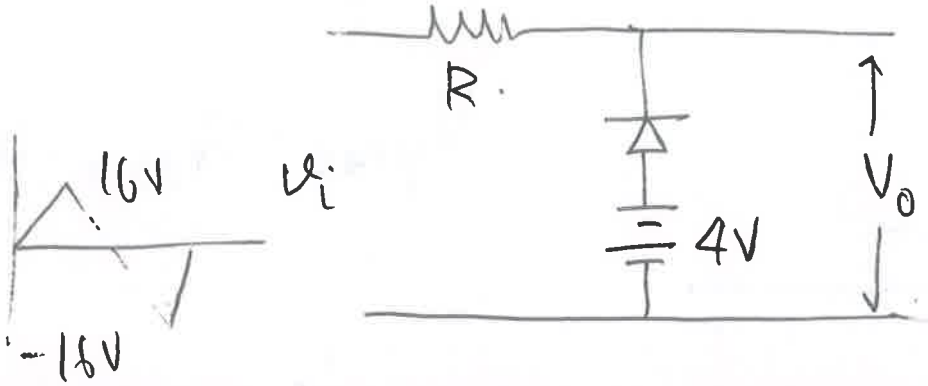

## Parallel clipper:-

When the diode in the clipper circuit is connected in parallel to the load then the circuit is said to be parallel clipper.



eg: 123 Determine $V_o$ for the network below.




S1: $V_o$ is across diode in reverse polarity and a battery in +ve polarity of $V_o$.

$\Rightarrow$ When D-Conducts, $V_o = +4V$.

S2:

→ +ve cycle of $v_i$ – RB the diode.

→ –ve " " – FB " ".

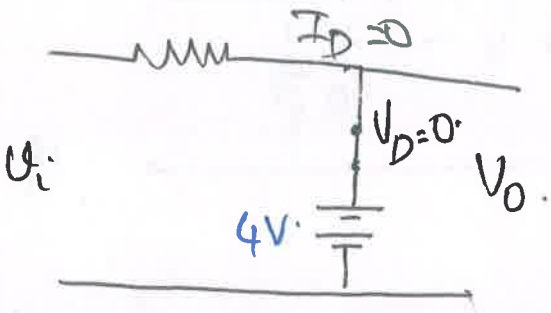
– dc battery – FB the diode leading to $V_o = 4V$.

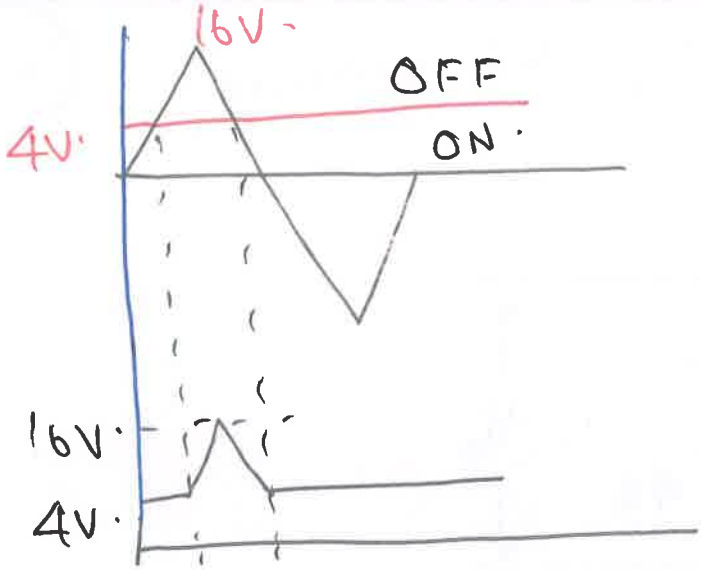
→ Normally when $v_i = 0$, D-ON – $V_o = 4V$.

When $v_i$ ↑ (+ve div), when $v_i = 4V$ – D-OFF

S3: Transition:

$v_i = 4V$.



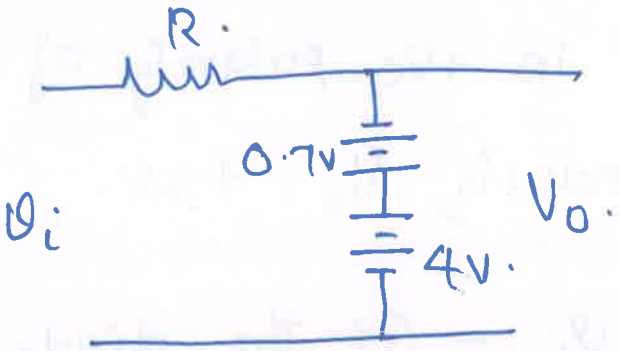



S4: At SS,

$V_o = 4V.$

$V_{opeak} = V_{ipeak} = 16V.$

eg: 124. Repeat 123 with practical diode.



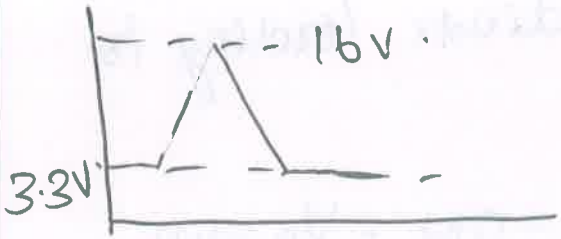

At SS: $V_o = 4 - 0.7$

$V_o = 3.3V.$

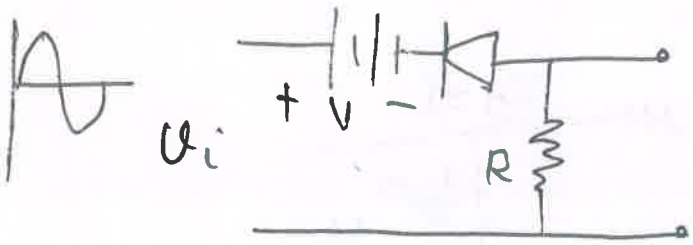
At transition.

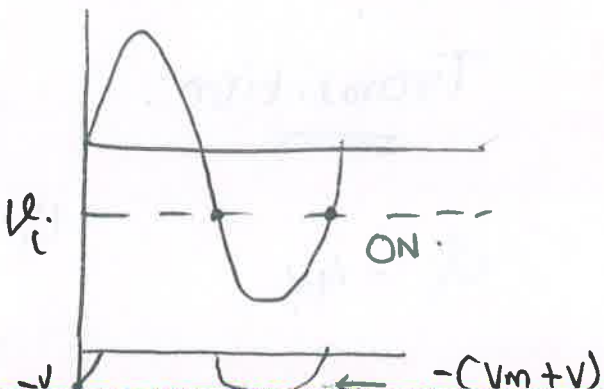
$V_i + 0.7 - 4V = 0.$

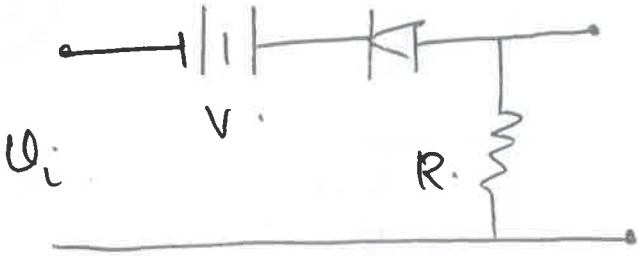
$V_i = 3.3V.$

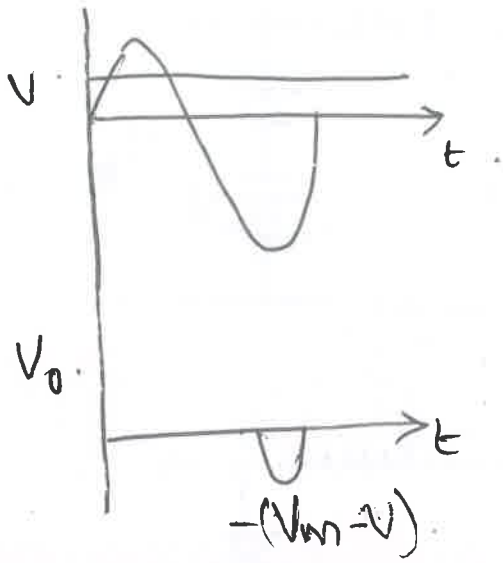


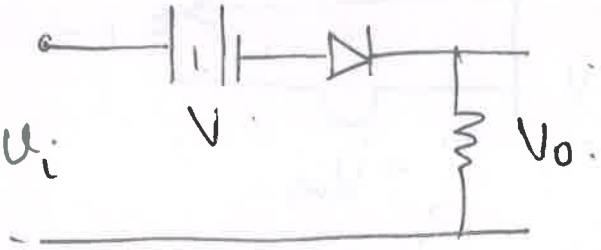

Other Series clippers:

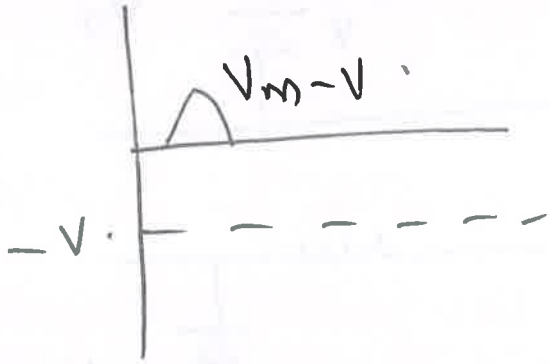


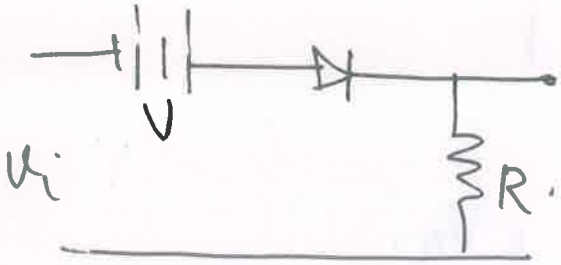



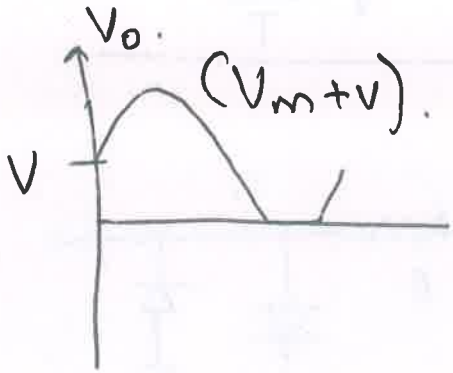


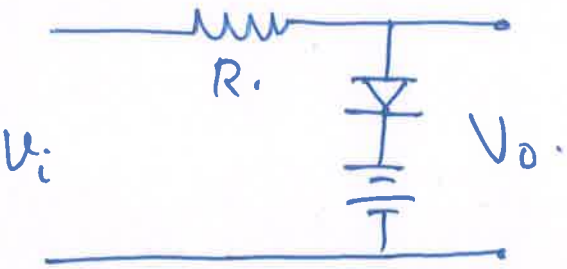



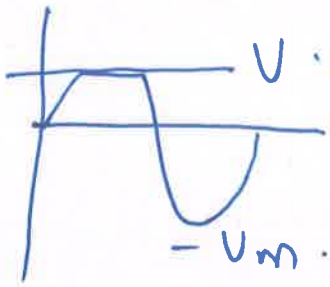



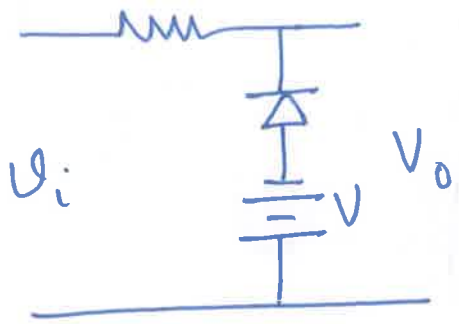



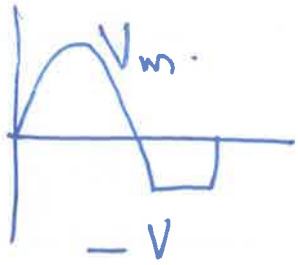



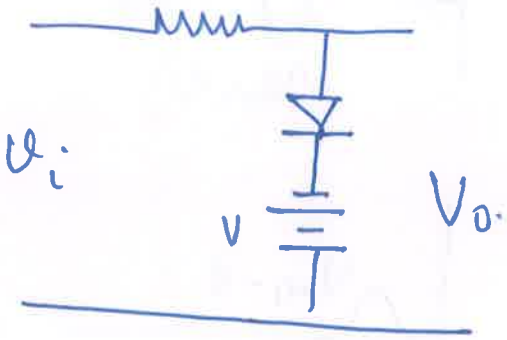

## Biased parallel clippers:

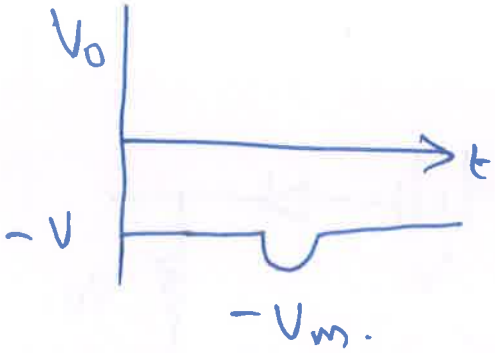


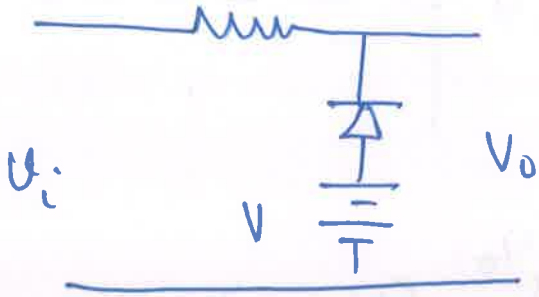



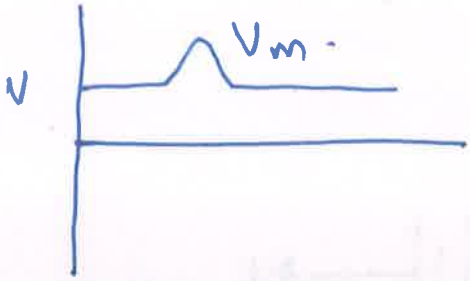

$v_i$
$V_0$
$V$

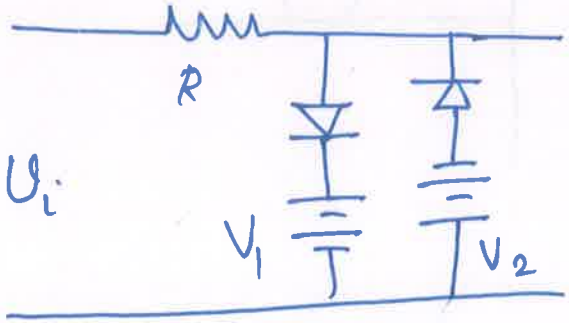

$V_m$.
$-V$

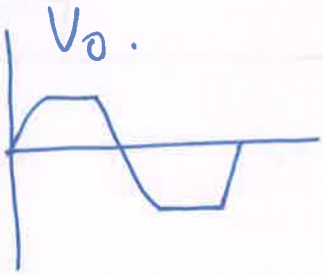

$v_i$
$V$
$V_0$.


$V_0$
$t$
$-V$
$-V_m$.


$v_i$
$V$
$V_0$


$V$
$V_m$.

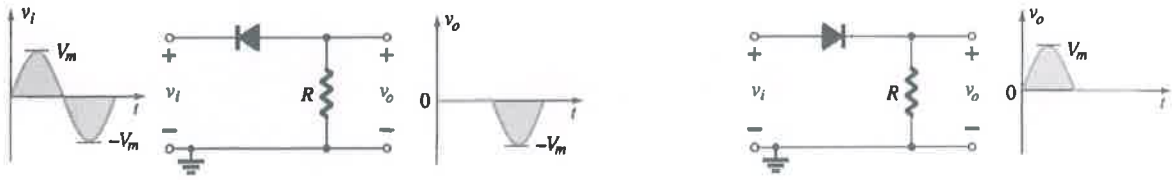

$R$
$v_i$
$V_1$
$V_2$


$V_0$.

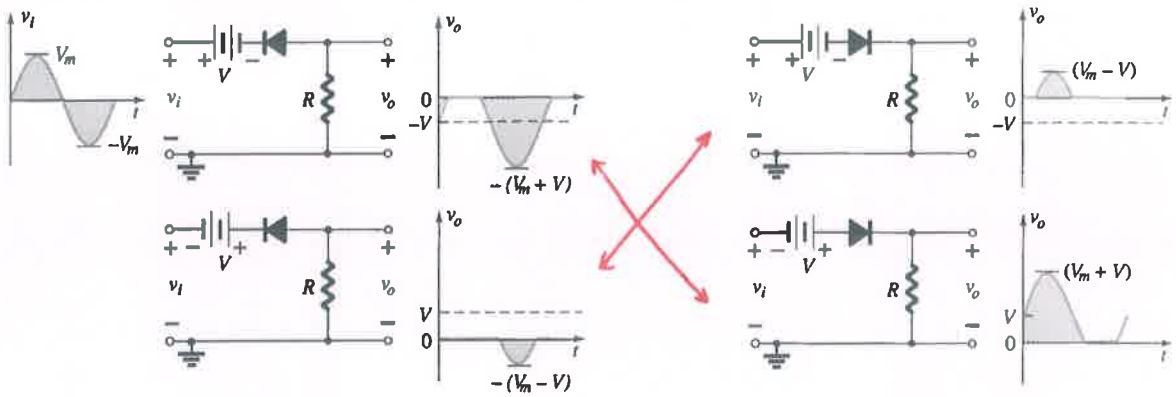
Simple Series Clippers (Ideal Diodes)

POSITIVE

NEGATIVE




Biased Series Clippers (Ideal Diodes)

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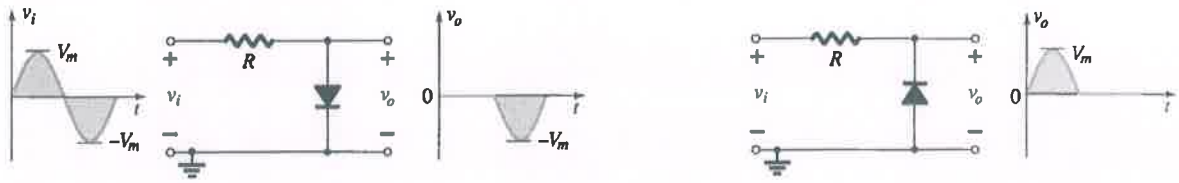
Simple Parallel Clippers (Ideal Diodes)




Biased Parallel Clippers (Ideal Diodes)

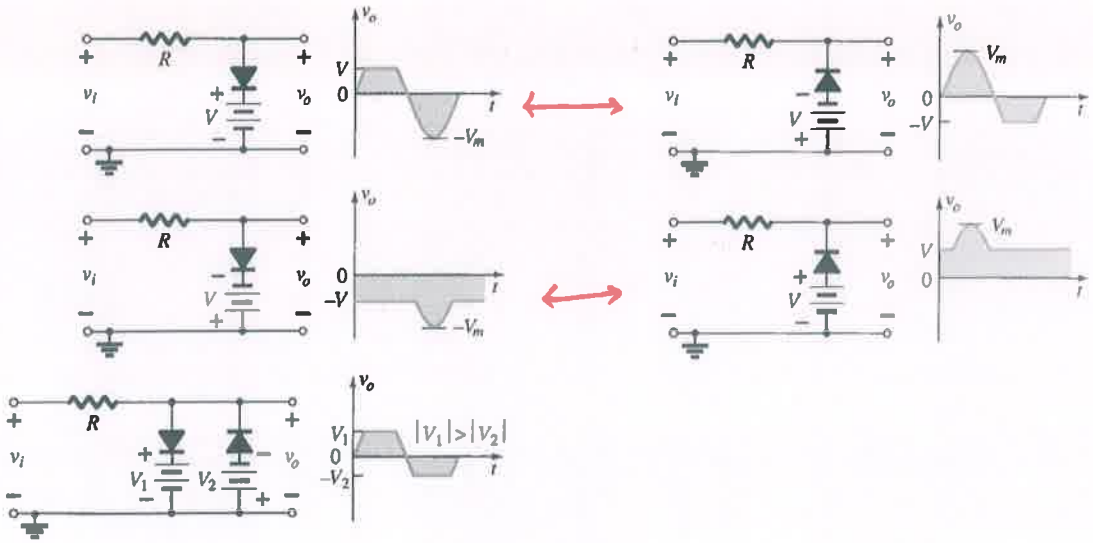



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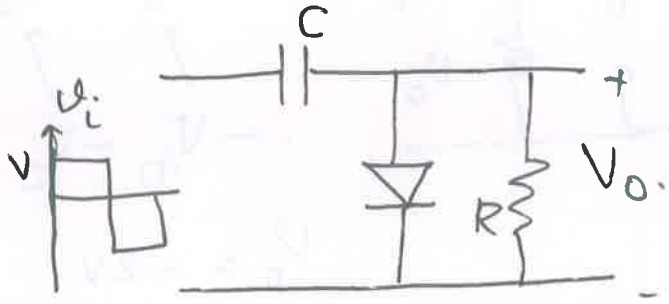
**FIG. 2.88**
*Clipping circuits.*

## Clampers:

Clamper consists of diode, resistor and capacitor. Clamper shifts the ac signal by a dc level.

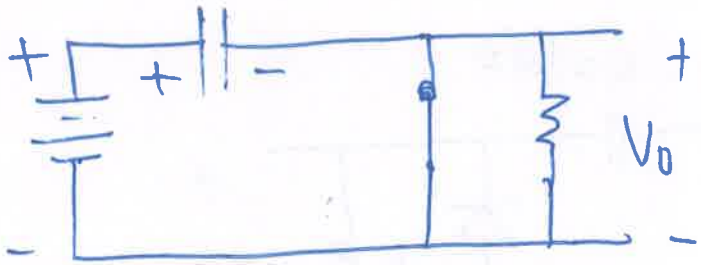
The time $\tau = RC$ must be large enough, so that 'c' doesn't discharge quickly.




## Steps to Solve:-

**Step 1:** Start analysis from the portion of $v_i$ that makes diode - FB.

**Step 2:** Once diode is FB, assume, c to be charged instantly.



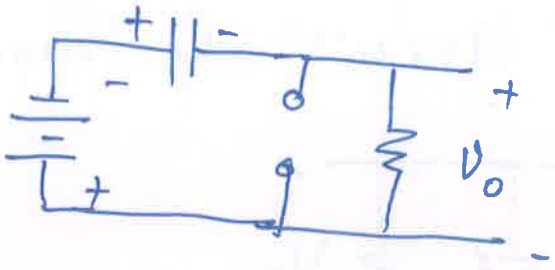

$\tau = RC$.

$\tau = 0$

[Charging of C is instantaneous].

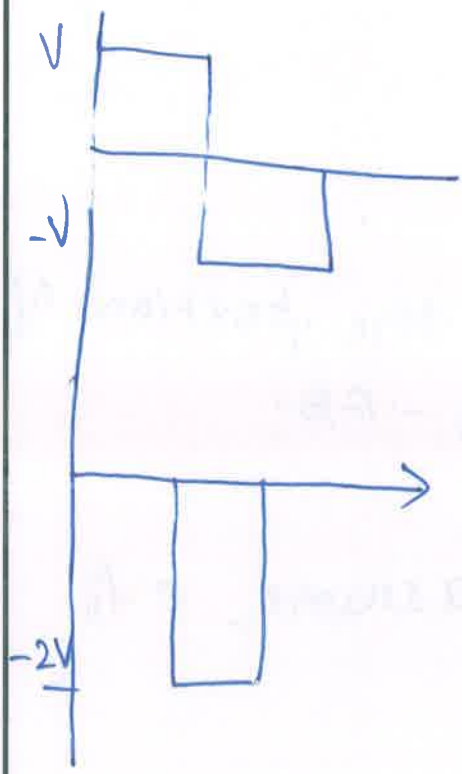
**Step 3:** Assume that during the period when diode is RB, 'C' holds on to its value.

**Step 4:** Keep a note on the polarity of $V_o$ throughout the analysis.

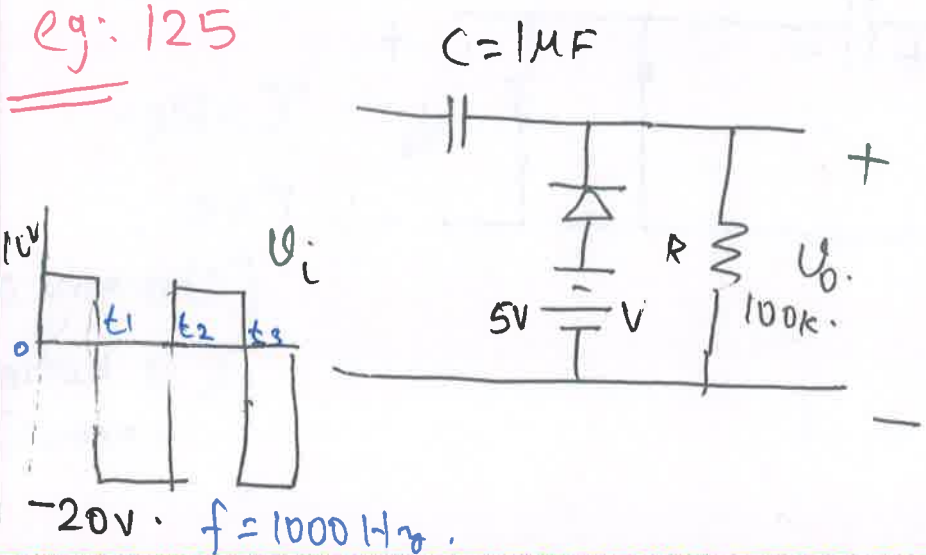



$$-\overset{V}{V_{in}} - \overset{V}{V_C} - V_o = 0.$$

$$V_o = -2V.$$

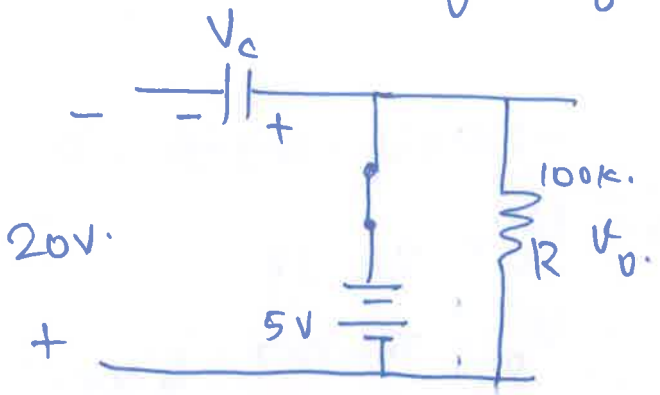



**eg: 125**



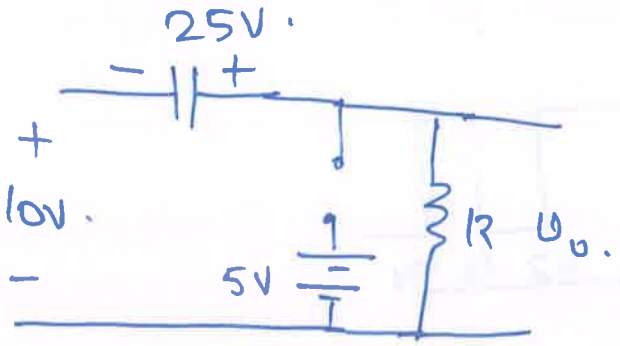

f = 1000 Hz.

S1: Start analysis from $t_1 \to t_2$.




$$-20 + V_c - 5V = 0.$$

$$V_c = 25V.$$

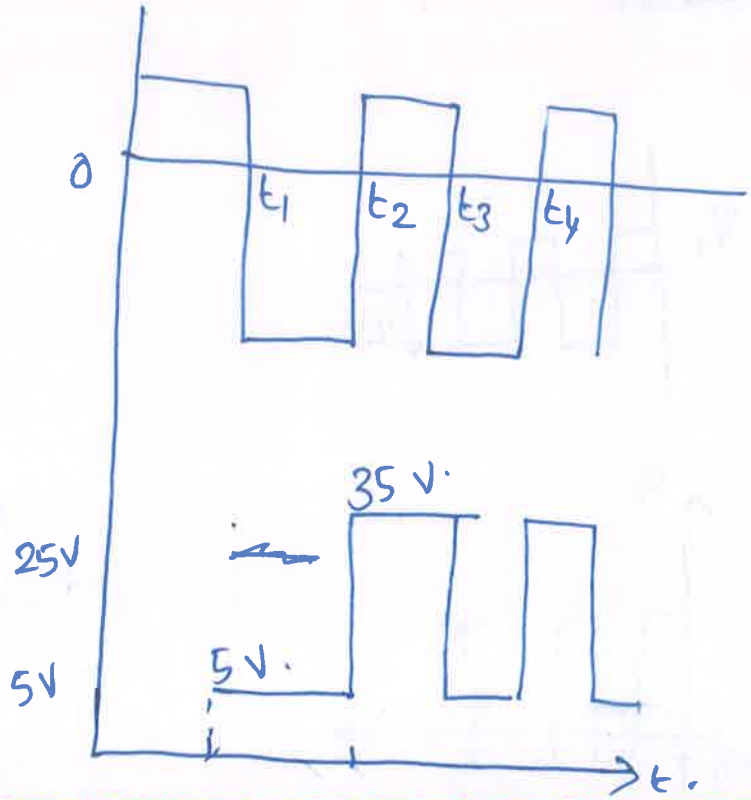



$$10 + 25 - v_o = 0.$$

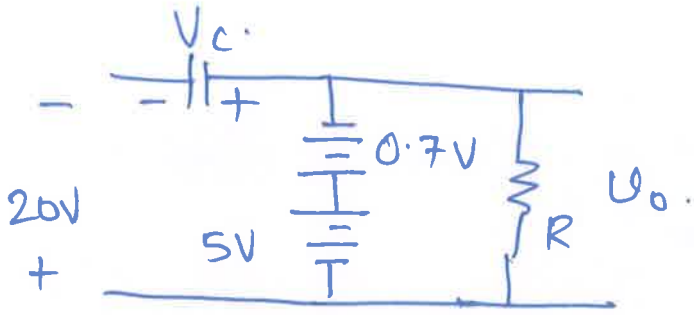
$$v_o = 35V.$$

$$\tau = RC = 100k \times 0.1\mu F = 0.01s = 10ms.$$

$$5\tau = 50ms.$$



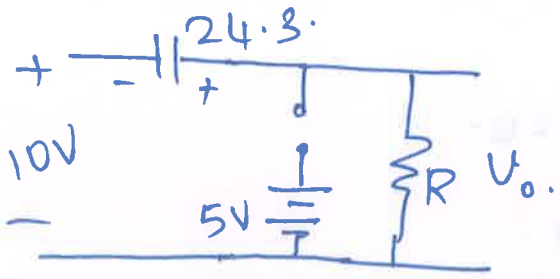
eg: 126: Repeat 125 with practical diode.




$-20 + V_c + 0.7 - 5 = 0.$

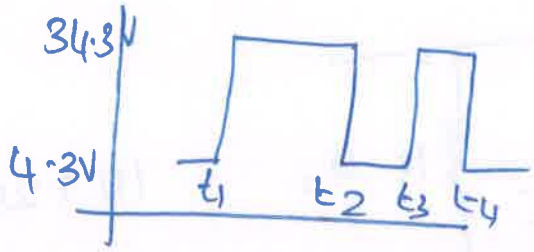
$V_c = 24.3V.$

$V_0 = 5 - 0.7 = 4.3V.$



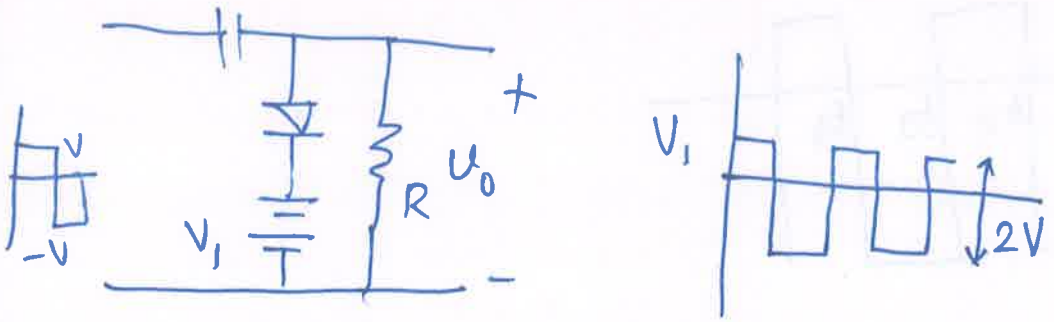

$10 + 24.3 - V_0 = 0.$

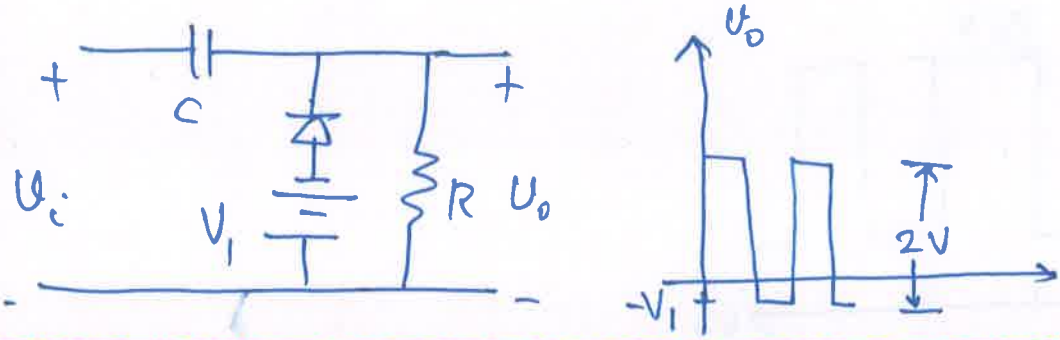
$V_0 = 34.3V.$




For more problems, refer the text.

Additional clamper circuits:-



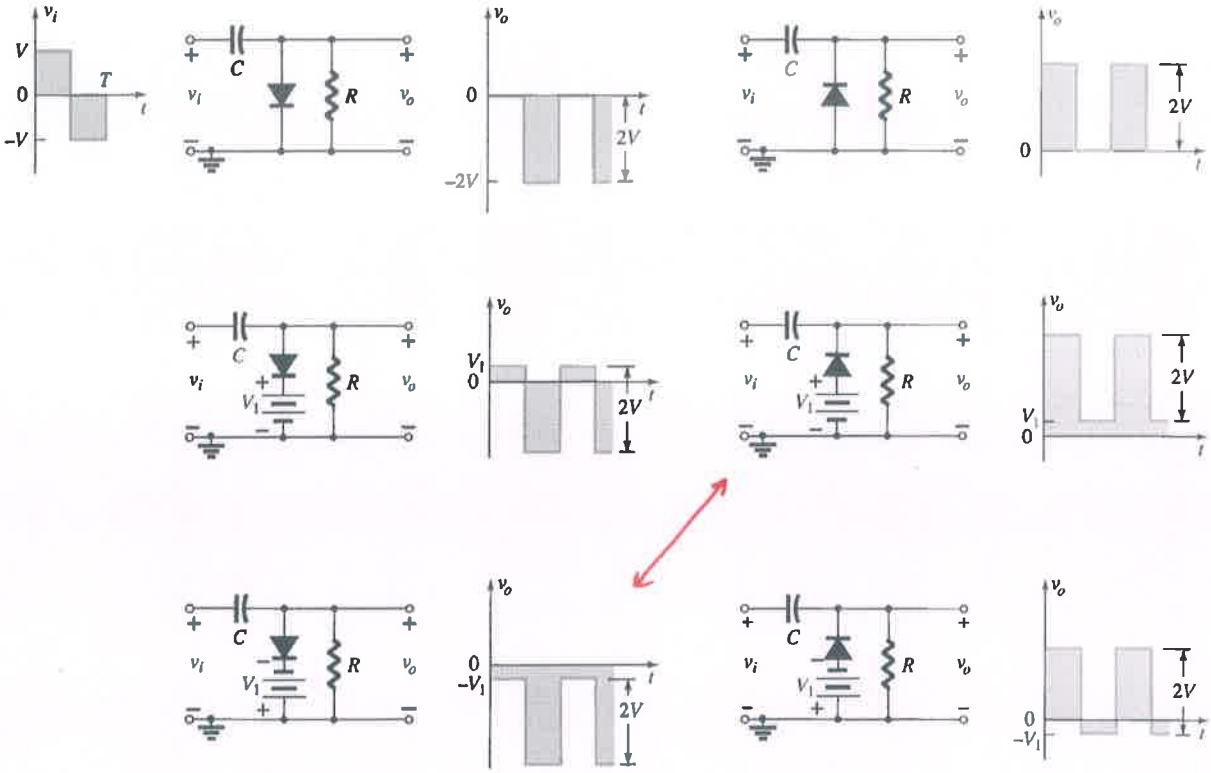



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**FIG. 2.100**
*Clamping circuits with ideal diodes ($5\tau = 5RC \gg T/2$).*

A number of clamping circuits and their effect on the input signal are shown in Fig. 2.100. Although all the waveforms appearing in Fig. 2.100 are square waves, clamping networks work equally well for sinusoidal signals. In fact, one approach to the analysis of clamping networks with sinusoidal inputs is to replace the sinusoidal signal by a square wave of the same peak values. The resulting output will then form an envelope for the sinusoidal response as shown in Fig. 2.101 for a network appearing in the bottom right of Fig. 2.100.

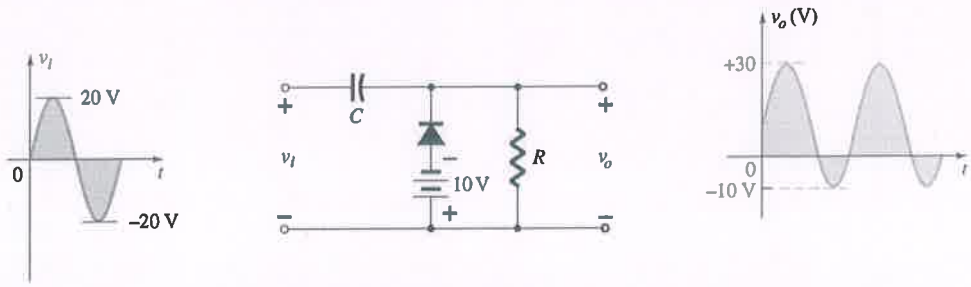
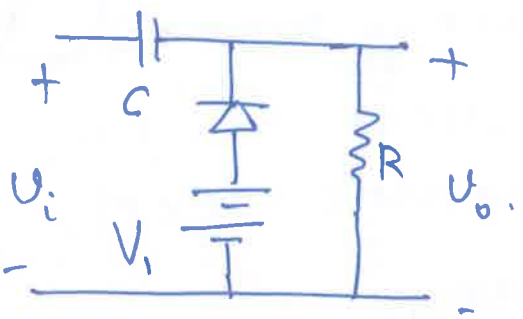
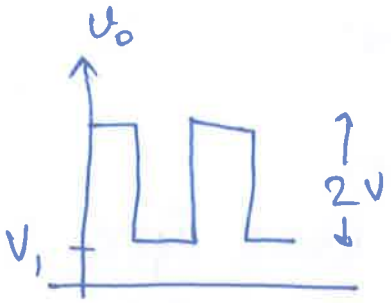


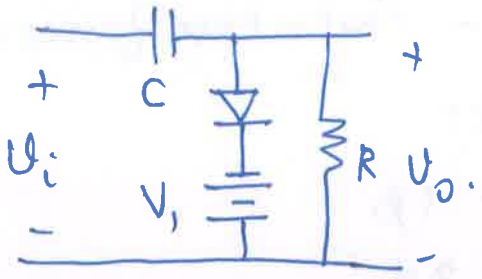

**FIG. 2.101**
*Clamping network with a sinusoidal input.*

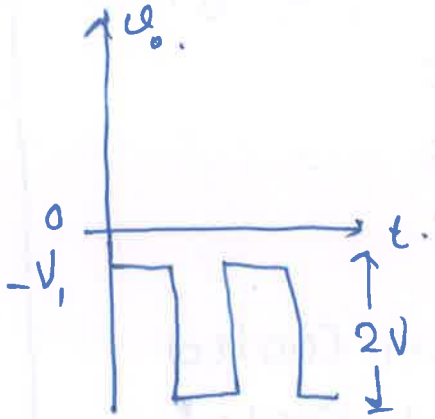
## 2.10 NETWORKS WITH A DC AND AC SOURCE

The analysis thus far has been limited to circuits with a single dc, ac, or square wave input. This section will expand that analysis to include both an ac and a dc source in the same configuration. In Fig. 2.102 the simplest of two-source networks has been constructed.




## Voltage doubler circuit:

Refer ppt.

## Electrical machines:

- Rotating eg. transformer
- Non rotating. eg. Motors, Generators.

## Transformer:

- Transformers transform one voltage level to the other voltage level w/o change in frequency.

### Classification:

- Step up
- Step down.

- distribution
- power

Construction :

- Core type
- Shell type
- Berry type

D :

- AF
- RF

Cooling :

- Air cooled
- Oil cooled

Measuring parameters

- Voltage (PT)
- Current (CT)

Windings :

- Autotransformers

Phases :

- $1\phi$
- $3\phi$.

$3\phi$ transformer connection

$\Delta - Y$

$Y - \Delta$

$Y - Y$

$\Delta - \Delta$.

Transformer parts :-

- Core
- Winding.

Core : It consists of Lamination, section include, L, T, I, F sections.

Winding :

- pry
- secy.

Turns ratio :

$$\frac{N_2}{N_1} = K > 1 \quad \text{step up transformer}$$

$$< 1 \quad \text{" down "}$$

Emf equation of transformer:-

$$E = 4.44 f \phi N \text{ Volts.}$$

losses inside transformer:.

- Constant - Core losses / Iron losses.
- Variable - Copper loss $(I^2R) \nearrow$.

Hysterisis eddy Current

$$\eta = \frac{\text{o/p power}}{\text{i/p power}} \Rightarrow \frac{\text{o/p}}{\text{o/p + losses}}$$

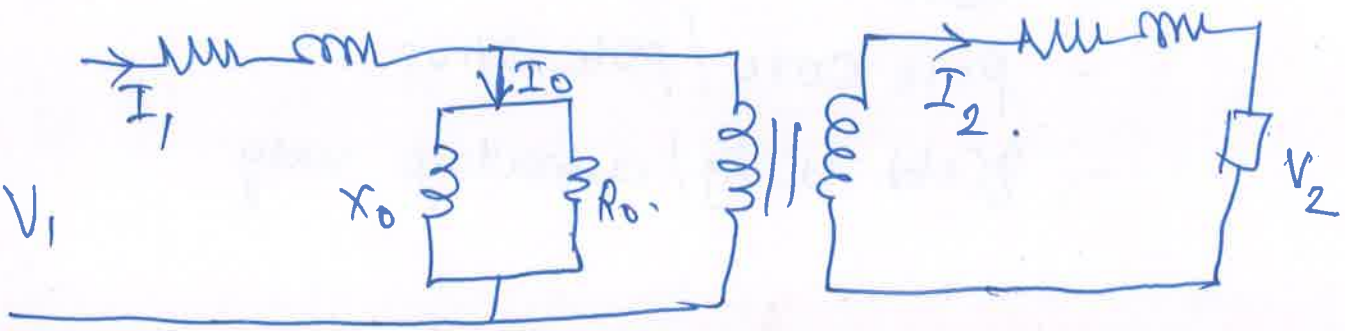
$$\text{o/p} = VI\cos\theta.$$

$$\eta = \frac{(VI)\cos\theta}{VI\cos\theta + P_i + I^2R}.$$

$$\eta \approx 95\%.$$

Another efficiency: All day efficiency.

Equivalent Circuit of transformer:-



Tests on transformer:-

- OC test
- SC test.

Machines (rotational).

- Motors
- Generators

Working of DC m/c:-

- DC m/c
- AC m/c.

DC motor/Generator:-

- Separately excited
- Self excited.
  - Series
  - Shunt
  - Compound
    - differential
    - Cumulative.

Parts of a dc motor:-

- Core
- pole core / pole shoe.
- field wdg / armature wdg

- Commutator
- brushes

## Emf in motor / generator:-

In generator - Induced emf
" motor - back emf

$$E = \frac{NP\phi Z}{60A}$$

N - speed, P = poles, $\phi$ - flux,
Z - Conductors, A = P - lap wdg
= 2 - wave ".
No. of parallel paths.

## AC motors / generators:

AC generators - Alternators.
" motors - Induction motors
- Synchronous motors
Squirrel Cage IM.
Slipring IM.
$S\phi$ = c start. c run. c start/run.
$3\phi$ = reluctance.

## Stepper motors:-

- PM SM.
- VR "
- Hybrid SM.

## Application examples:

- Car wiper s/m.
- CD disk drive, Motherboard cooling fan.

Working of SM:

- Single coil excitation.
- Double " "
- Half step "